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MILITARY SPECIFICATION

TESTING, STORE SUSPENSION AND RELEASE EQUIPMENT, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope. This specification covers laboratory tests to be used in determining the suitability of all types of store suspension equipment intended for military aircraft use.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standard (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-E-6051	Electromagnetic Compatibility Requirements, System Standards

MILITARY

MIL-STD-461	Electromagnetic Emissions and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-704	Aircraft Electric Power Characteristics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requirements Department, Code SR3, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 6635

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MIL-STD-1385 Preclusion of Ordnance Hazards in Electromagnetic Fields, General Requirements for

HANDBOOKS

MILITARY

MIL-HDBK-235-1, -2, -3 Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems and Systems

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications The following document(s) form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 Standard Test Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 Test condition tolerance. Tolerance is defined as the range of indicated values taken from test instruments having errors as specified in 3.2. Unless otherwise specified, test condition tolerances shall be as follows, exclusive of thermocouple accuracy:

a. Tolerances for use with test chambers·

1 Air temperature at the control sensor· ±2 8°C (±5°F) in the range of -73 3° to 93 3°C (-100° to 200°F). For other temperatures, the tolerance shall be ±3 percent of the indicated value of temperature in °C

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2. Pressure. When measured by devices such as manometers, ±5 percent or ±1 5 mm (0.059 inches) of mercury, whichever provides the greater accuracy. When measured by devices such as ion gages, ±10 percent to 10^{-8} Torr.

3. Relative humidity at the control sensor: ±5 percent.

b. Other tolerances:

1.	Vibration amplitude:	Sinusoidal, ±10 percent.
2.	Vibration frequency:	±2 percent, or ±1/2 Hz below 20 Hz.
3.	Acceleration or shock:	±10 percent.
4.	Dive or climb attitude	±3 degrees
5.	Torque:	±5 percent.
6	Distance	±1/32 inch

3.2 Error of test instruments. The error (inaccuracy) of test instruments shall be equal to or better than the tolerances specified in 3 1

3.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions

a. Test temperature. Unless otherwise specified, the test temperature, hereafter designated room temperature, shall be 25° ± 10°C (77° ± 18°F). At other temperatures specified, the tolerance shall be ±2.8°C (±5°F). All parts of the item shall be stabilized at the specified temperature prior to conducting any tests Unless otherwise specified, temperature stabilization will have been attained when the temperature of the surface of the largest mass of the item does not change more than ±2 8°C (±5°F) per hour

b. Test voltage. Unless otherwise specified, the nominal test voltages, at the item terminals or connectors, shall be 115 ± 10 volts root mean square (rms) line-to-neutral for the 115/200 volt alternating current (Vac) system and 28 ± 2 volts line-to-ground for the 28 volt direct current (Vdc) system. However, the item shall operate satisfactorily between the voltage limits as specified in MIL-STD-704. Where an electrical pulse is specified herein, the tolerance of the duration of the pulse shall be ±10 percent or 0.002 second, whichever is greater.

c. Test attitude Unless otherwise specified, the test shall be conducted in an attitude simulating horizontal level flight, with the item mounted in a manner similar to that used in mounting the item in the aircraft

d. Sway braces. Unless otherwise specified by the acquiring activity, release tests on the item shall be conducted with the sway braces adjusted to preload values specified in the item's

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service manual, or item detailed specification, or as directed by the acquiring activity.

3 4 Auxiliary equipment. When the item is operated in conjunction with separate, stand-alone mechanisms such as inflight locks, electric fuzes, arming units, or solenoid releases, tests shall be conducted with prequalified mechanisms only. The acquiring activity, however, reserves the right to permit the use of unqualified mechanisms All tests of section 4 pertaining to detent style/spring loaded ball type arming units shall be deleted if the store suspension and release equipment under test has zero retention force (ZRF) arming units However, the deleted tests of section 4 shall be replaced by appropriate tests and functional operations for ZRF arming units. The ZRF arming unit functional operations shall be expanded to include all tests of table I It is the responsibility of the acquiring activity, via the written test procedure of 3 7, to define the ZRF arming unit tests such that proper function of the arming units is verified under all service conditions. The contractor is responsible when conducting the tests of table I to ensure that

a Each stand-alone mechanism is evaluated in accordance with its detail specification

b That each unique feature of the suspension equipment (an inflight lock for example) is evaluated in accordance with the suspension equipment's detail specification or per acquiring activity direction.

Stand-alone mechanisms and other unique features of the suspension equipment shall be tested and evaluated per acquiring activities direction even if not specifically required in section 4 (see 3.6 and 3.7)

3 5 Performance requirements The item shall operate satisfactorily when subjected to the environments in table I as detailed herein (see 6 3) The acquiring activity reserves the right to delete tests and alter the test sequence in table I.

TABLE I. Tests and sequence

Test	Requirement paragraph	Number of items								
		Two		Three			Four			
		A	B	A	B	C	A	B	C	D
1. Physical characteristics	4.2	1	1	1	1	1	1	1	1	1
2. Mechanical	4.3	3		3			3			
3. Electrical	4.4	2	2 1/	2	2 1/	2 1/	2	2 1/	2 1/	2
4. Lag time	4.5	4		4			4			
5. Icing	4 6	6		6			6			
6 Vibration	4.7		5		4			4		
7. Shock	4.8		4		3			3		
8. High-g	4.9	8		8			8			
9. Life	4 10	9		9			9			
10. Ejector system	4 11		14	11				9		
11 Sand	4 12	7		7			7			

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TABLE I Tests and sequence - Continued.

Test	Requirement paragraph	Number of items								
		Two		Three			Four			
		A	B	A	B	C	A	B	C	D
12. Salt 1/	4.13	-	-	-	-	-	-	-	-	-
13 Static	4 16		13	10				8		
14 Rain	4 17		6		5			5		3
15 Humidity	4.18		9			5			3	
16. Fungus	4 19		10		7				4	
17 Altitude	4 20		11			4			5	
18 Sunshine	4 21		7		6			6		
19. Temperature shock	4.22		8		8			7		
20. Lock shut	4.23	13				9			9	
21 EMC	4.24		3			3			3	
22 Stress corr	4 14	10				6			6	
23. Salt alt	4 15		12		9					4
24 Sympathetic ignition	4 25	5		5			5			
25. Contamination	4.26	11				7			7	
26. Chemical hardening	4.27	12				8			8	

1/ The salt spray test specified in 4.13 will be performed on non-structural items only, such as arming units, release units, and electrical assemblies, unless the acquiring activity directs this test to be substituted for the stress corrosion test (4 14) on the test items

2/ Paragraph 4 4.2.2 only.

3 6 Performance criteria. The item shall operate to safely release/eject stores during or after each environment without serious degradation of functional parameters. Parameters normally considered (because they can be defined by direct measurement) are electrical continuity and resistance, lag time, breech and ejector cylinder pressures, store ejection velocity, store pitch rate, store ejection acceleration, ejection force, transfer time, release torque, or latching torque, for example. However, there are other functional parameters which define the performance of the item. Some examples of these parameters are the operation of an inflight lock during vibration, the proper capture of an arming wire by an arming unit to arm a store during high-g release, or the proper operation of a release solenoid during icing tests All parameters that must be measured or operated during testing to ensure proper item operation under service conditions cannot possibly be described in section 4. Therefore, it is the responsibility of the acquiring activity to include functional parameters not specified in section 4 and deemed critical to proper item operation in the written test procedure of 3.7. Measurable functional parameters shall be described in the item's detailed specification. If these parameters are not in the detailed specification, initial functional and receiving tests shall be utilized to obtain data which shall serve as the comparison base for all subsequent tests, or an additional test may be specified by the acquiring activity to establish initial base line data Consistency of operation shall be ±5 percent of the

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initially measured base l ne data condition These tolerances snall alsc apply to parameters listed in the item's detailed specification if no tolerances are given. When functional parameters are not specified in the item's detailed specification, the parameters obtained from the initial functional and receiving tests shall indicate a degree of consistency between all items received for test For example, if two items are received, one should not have a release torque of 10 inch-pounds while the second has a release torque of 100 inch-pounds, even if these values remain constant throughout all testing

3.7 Test procedure Prior to conducting any tests, there shall be a written test procedure (see 6 3) The purpose of the test procedure is to clarify, add to or delete from each test requirement of this specification so that the tests are conducted to demonstrate the ability of the item and all of the item's unique features to withstand its service environment. The contractor shall propose the data measurement methods, including tolerances, for all performance criteria in the test procedure

4 QUALITY ASSURANCE PROVISIONS

NOTE

Each test method oescribed in the following paragraphs is divided into three distinct sections The first section is a statement about the test, the second is the exact test(s) to be conducted and the third is the failure criteria for the conducted tests No test of this section shall be conducted without a written test procedure Attention is called to section 3, particularly 3.4, 3 6, and 3 7

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

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4.2 Physical characteristics

4.2.1 Conformance The item shall be inspected to determine conformance to the drawings for the item being procured Features/mechanisms of the item shall meet the requirements of 3.4, 3 6, and 3.7.

4.2 2 Procedure

4.2.2 1 Weight Each item shall be weighed in its flight condition Devices such as safety pins, cartridges, and ground servicing equipment, shall not be included in the weighing.

4.2.2 2 Fit The ability of each item to suspend the various bombs, missiles, mines, torpedoes, and fuel tanks specified for the item shall be verified as directed by the acquiring activity In addition, the item shall accept the cartridges which are specified, shall be capable of sway bracing, and shall fit into pylons and adapters Specified hoisting and loading equipment shall be checked for interference by attaching specified stores to the item

4 2 2 3 Visual The item shall be visually inspected for workmanship to ensure, within design limitations, the ability of the item to meet its performance requirements

4 2 3 Failure criteria If the item does not meet the following criteria, it shall be cause for failure

4.2.3.1 Weight The weight of the item shall be within the limits specified. If no weight is specified, the weight shall be measured and reported to the acquiring activity.

4.2.3.2 Fit. The item shall be capable of suspending the stores specified. Hoisting and loading equipment shall not interfere with the attachment of the stores. The item shall fit specified pylons or adapters to verify that item/store assemblies can be attached and maintenance or service functions performed.

4.2 3.3 Visual The item shall not exhibit signs of unauthorized repair, welding, heavy burrs, or parts assembled by introduction of high stresses not prescribed in the drawings since these are typical signs of inferior workmanship and are unacceptable. Decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, shall not be peeling, chipped, loose, or broken. Manufacture, finish, and assembly shall be as specified.

4.3 Manual function tests (see 6.4.11).

4.3 1 Operation. The item shall be tested to ensure that it manually functions as required Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3 7

a Inflight operation The item shall be subjected to the tests of 4.3 2 as specified.

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b Ground service operation

1. Effect of sway braces All releases shall be performed with the sway braces of the item adjusted to the specified preload values If the item fails to release the required stores, the sway brace preload shall be reduced incrementally and releases shall again be attempted until the store(s) is released The effect of sway bracing shall be reported to the acquiring activity

2 Operation The item shall be subjected to the tests of 4 3 2, except 4 3 2 1.2 and 4 3 2.1 3, as specified

4 3 2 Procedure

4 3 2.1 Operating forces, manual release mechanism

4 3 2.1 1 Horizontal release Three releases of the lightest and three releases of the heaviest store shall be made under each of the following item stabilized conditions

a 93 3°C (200°F)

b Room temperature.

c -56 7°C (-70°F)

The torque required and the distances that the controls must travel under each condition to attain the release point and full travel of the release mechanism shall be measured

4.3 2 1.2 Dive and climb release, horizontal bombing items Three releases of the lightest and three releases of the heaviest store shall be made at room temperature only with the item in each of the following attitudes:

a. 15° climb

b. 15° dive

The torque required and the distances that the controls must travel under each condition to attain the release point and full travel of the release mechanism shall be measured

4.3.2.1.3 Dive and climb release, dive bombing items Three releases of the lightest and three releases of the heaviest store shall be made at room temperature only with the item in each of the following attitudes:

a. 90° climb

b. 70° dive

The torque required and the distances that the controls must travel under each condition to attain the release point and full travel of the release mechanism shall be measured

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4.3.2.2 Operating forces, mechanical arming units The arming unit may be removed from the item to conduct testing if this does not affect the arming function. Applied voltage shall be as specified.

4.3.2.2 1 Arming wire pulls, safe setting. With the mechanical arming unit electrically set SAFE, the force that must be exerted to pull all compatible arming wires out of each unit shall be measured The pulls shall be made parallel with the side of the item (in the direction of the store release); straight down, 45° forward and aft; and 30° to each side in the transverse direction (see figure 1). When some part of the item interferes with the arming wire pull in a specified direction, this direction of pull shall not be accomplished and the interference shall be reported to the acquiring activity. The tests shall be conducted three times in each direction of pull and under the following item stabilized conditions

a. 93.3°C (200°F).

b. Room temperature.

c. -56.7°C (-70°F)

4.3 2 2 2 Proof load, armed setting With the mechanical arming unit electrically ARMED, a proof load of 68 04 +2 27, -0 kilograms (150 +5, -0 pounds) shall be applied per figure 1, direction 1, for all units

The tests shall be conducted under the following item stabilized conditions

a. 93.3°C (200°F).

b. Room temperature.

c. -56.7°C (-70°F)

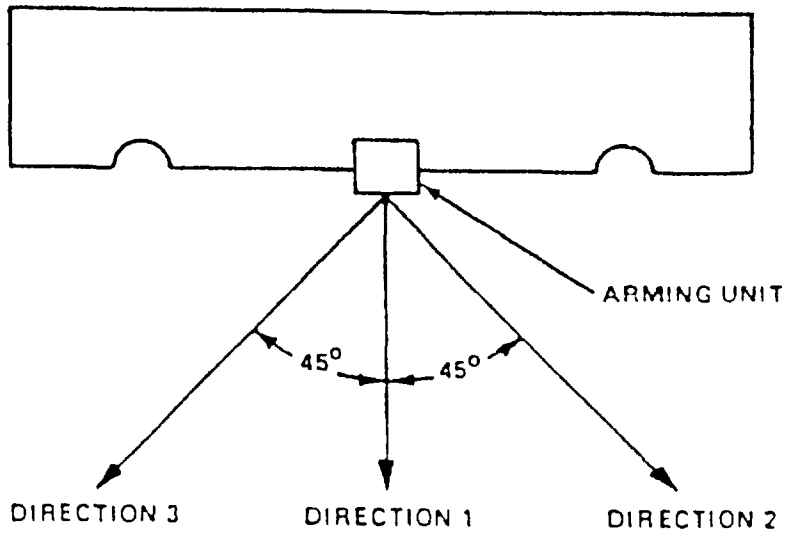
4.3.2.3 Electric fuze arming The requirements for the electric fuze arming unit tests shall be as specified Lanyard clips, microswitches, pull-out forces, clearance, and safety wiring, attachment to the item shall be inspected or functionally evaluated as applicable

4.3.2.4 Visual. The item shall be visually inspected after each of the conditions above to ensure that it has not been damaged due to the manual functioning tests.

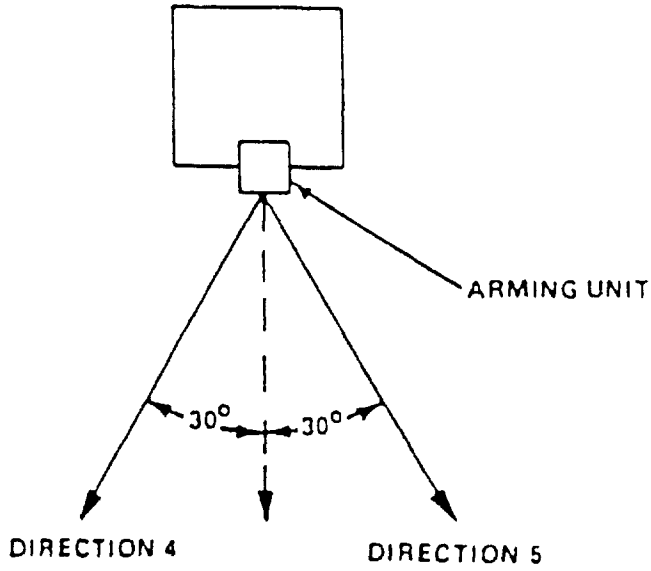
4.3.3 Failure criteria. If the item does not meet the following criteria, it shall be cause for failure.

4.3.3.1 Operating forces, manual release mechanism The item shall release in any of the attitudes or under any of the temperature conditions specified. The operating forces or distances shall be as specified or shall be in accordance with 3.6. There shall be no damage to the item as a result of store release.

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PULL DIRECTIONS IN FORE AND AFT AXIS



PULL DIRECTIONS IN TRANSVERSE AXIS

FIGURE 1. Directions for arming wire pulls. 1/

1/ Use figure 1 for detent style/spring loaded ball type arming units only. For zero retention force arming units see 3.4.

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4 3 3.2 Operating forces, arming units The arming unit shall hold the 68 04 kilogram (150 pound) proof load (armed) without any damage. It shall function after load applicatior The arming unit shall release between the minimum and maximum loads for SAFE release specified in its detailed specification without any damage It shall function after any SAFE release If SAFE release loads are not available, see 3 6.

4 3.3.3 Electric fuze arming The electric fuze arming unit shall meet the requirements specified.

4.3.3.4 Visual. The item shall not malfunction as a result of excessive wear (except replaceable parts), nor shall the testing cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, or discolor There shall be no damage to the item as a result of store release.

4 4 Electrical characteristics.

4 4 1 Electrical components. The item shall be tested to ensure that it electrically functions as required Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3.7 Any of the electrical components may be removed from the item for purposes of checking the component's electrical characteristics provided such removal does not compromise the electrical integrity of the item. For example, wires may be mechanically disconnected from junctions, but shall not be unsoldered or cut

4 4.2 Procedure

4 4.2 1 Store sensing switch. All prescribed switching functions of the store sensing switches shall be checked for satisfactory operation, both with and without appropriate stores installed. The electrical continuity and mechanical characteristics; i.e., stiffness of spring, shall be inspected and the attachments evaluated.

4.4.2 2 Operational test at rated voltage. To ensure that the item has been properly assembled, all circuits shall be operational when a specified voltage is applied to the item terminal(s) or connector(s). Tested circuits can include, but shall not be limited to, the safety switch, actuators (LEMA, IFOBRL), stepper switch, release circuits, arming units, and CBU circuits. The current applied to each circuit shall be the nominal value for normal operation. Current limiting devices, such as resistors, shall be used to ensure proper current flow in the various test circuits.

4.4.2.3 Maximum temperature. The maximum temperatures for insulation used in electromagnetic units shall be obtained by utilizing one of the following methods

NOTE

Methods of temperature measurement will be left to the discretion of the testing activity. If any disagreement arises as to the class of wire insulation used in any of the electrical equipment, the acquiring activity will make the final decision. Test instrument accuracy shall each be within 1 percent of indicated value

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4.4.2 3.1 Continuous duty electromagnetic unit Continuous duty electromagnetic units are defined as those which, when energized continuously from the highest specified voltage source while in an ambient temperature of 25° ± 10°C (77° ± 18°F), will not exceed the maximum allowable temperature for the class of insulation used, as specified in table II. The test voltage shall be applied continuously until temperature stabilization is reached. The resistance method shall be used to determine the maximum temperature. The following formula shall be used:

$$\text{Maximum temperature } °C = (234.5 + T_c)\frac{R_h}{R_c} - (196\ 7 + T_a)$$

Where

R_c = cold resistance of coil

R_h = hot resistance of coil.

T_c = ambient temperature (°C) when cold resistance is measured

T_a = ambient temperature (°C) during last quarter of test

TABLE II Classes of insulation vs temperatures

Classes of Insulation	Maximum Temperatures	
	°C	°F
O	90	194
A	105	221
B	125	257
C	205	401
H	208	401

4.4.2.3.2 Intermittent duty electromagnetic unit. Intermittent duty electromagnetic units are defined as those which, when energized for 15 seconds from the nominal specified voltage source while in an ambient temperature of 25° ± 10°C (77° ± 18°F), will not exceed the maximum allowable temperature for the class of insulation used, as specified in table II In order to determine the change in resistance of the coil, an ammeter shall be used in series with the coil, the combination of which shall be energized from a constant electromotive force (emf) the same as the nominal specified voltage Readings shall be taken of the current every 5 seconds, for at least 15 seconds The following formula shall be used.

$$\text{Maximum temperature } °C = (234.5 + T_c)\frac{R_h}{R_c} - (210.5 + T_a)$$

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Where.

R_c = cold resistance of coil.

R_h = hot resistance of coil

T_c = ambient temperature (°C) when cold resistance is measured

T_a = ambient temperature (°C) at 15 seconds time

4 4.2.4 Insulation tests

4 4 2 4 1 Dielectric breakdown to ground. This test applies to all electrical circuits of the item where the ground lead is accessible and can be broken After breaking the ground lead, an alternating test current of 1000 ± 50 volts rms shall be applied between each electrical circuit and all other electrical circuits including ground. The frequency of the alternating current may be any value between 50 and 100 Hz and the shape of the wave shall be in conformance with good commercial practice The voltmeter method shall be used to measure the test voltage. The voltmeter shall derive its potential directly from the high voltage circuit through an auxiliary transformer The voltage shall be increased from zero to test voltage within one half minute and shall be maintained for a period of 1 +0 2, -0 minute. The current flow shall be monitored

4 4 2.4 2 Dielectric breakdown between layers. An emf of at least 166 percent of rated voltage shall be applied across any solenoid coil. This circuit shall be made and broken rapidly 10 times with a minimum "on" time of 0 5 second. The instruments used in this test shall be accurate within 1 percent of indicated value Any ambient temperature effects shall be taken into account when the resistance measurements are made.

4 4.2.4.3 Insulation resistance. This test applies to all electrical circuits of the item where the ground lead is accessible and can be broken. After breaking the ground lead, the item shall be subjected to the maximum temperature specified in the detailed specification or 93°C (200°F), if no temperature is specified, for a period not less than 4 hours The insulation resistance shall be measured by means of a constant potential "megger" or by the voltmeter method under a potential of 500 ± 50 Vdc. The item shall then be removed from the heat, and be allowed to cool in an atmosphere of 100 percent relative humidity until stabilized at room temperature, after which the insulation resistance shall be measured as described above

4.4.2.5 Minimum operating voltage, release and arming. The minimum operating voltage necessary to effect releases of the lightest and heaviest stores and the minimum voltage to effect positive operation of arming units, IFOBRLS, LEMAS, and secondary releases, shall be determined. The minimum voltage shall be determined by measuring with a voltmeter at the item connector. The meter shall have an accuracy of ±1 percent of the indicated value.

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4 4 2 6 Transfer switch (see 6 4 20) The time for the transfer switch to actuate shall be measured from the end of the release impulse until current in the transfer circuit reaches a steady state condition, free of switch chatter fluctuations. The tests shall be conducted at the lowest and highest specified voltage for the item under the following item stabilized conditions:

a 93°C (200°F).

b Room temperature

c -56 7°C (-70°F)

4 4 2 7 Electric fuze arming. The electrical requirements for the electric fuze arming unit tests shall be as specified

4.4.2.8 Visual. The item shall be visually inspected after each of the conditions above to ensure that it has not been damaged due to the electrical characteristics tests.

4 4 3 Failure criteria If the item does not meet the following criteria, it shall be cause for failure.

4 4 3.1 Store sensing switch The switch shall operate mechanically and electrically in the specified manner while loading and unloading stores The switch shall be securely fastened to the item so that loading and unloading the store shall cause no damage

4.4.3.2 Operational test at rated voltage Each circuit shall operate in the specified manner when actuated with the specified voltage. For example, safety switches shall not allow store release when in the SAFE position, arming units shall hold 68.04 kilograms (150 pounds) when actuated (see 4 3.2.2.2); stepper switches and intervalometers shall step to all selected positions when actuated. The electrical load or resistance in each circuit shall be as specified.

4.4.3.3 Maximum temperature The maximum permissible temperatures for various classes of insulation, when tested in accordance with 4.3.2.3, are listed in table II. Class A insulation shall be assumed unless otherwise directed by the acquiring activity.

4.4.3.4 Insulation tests

4 4.3.4.1 Dielectric breakdown to ground. Any indication of damage by arcing, or current flow in excess of 0 001 ampere, during the 1 minute test period shall be considered a failure.

4.4.3.4.2 Dielectric breakdown between layers. After being subjected to the specified voltage and after having returned to stabilized room temperature, the unit shall not show a change from the original resistance in excess of 1 percent of indicated value.

4.4.3.4.3 Insulation resistance. Under either temperature condition, a measured or calculated insulation resistance of less than 1 megaohm shall be cause for rejection of the item

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4 4 3 5 Minimum operating voltage, release and arming. The minimum voltage shall not exceed the minimum specified voltage for the particular unit or subassembly Inflight safeties, primary releases, arming units, and secondary releases shall all be capable of functioning as specified Arming units shall hold 68.04 kilograms (150 pounds) when energized (see 4 3 2 2 2)

4.4.3 6 Transfer switch Unless otherwise specified, the interval for the transfer switch to actuate shall not exceed 0 020 second

4.4.3 7 Electric fuze arming The electric fuze arming unit shall conform to its electrical requirements as specified

4.4.3.8 Visual The item shall not show signs of degradation (discoloration, and burnt insulation) as a result of any of the above tests

4.5 Lag time or time of release determination (see 6 4 7).

4 5.1 Operational parameters The item shall be tested to ensure that the operational parameters conform to requirements when subjected to store release Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3 7

4 5.2 Procedure

4.5 2 1 Horizontal release, minimum specified voltage, 0 020 second impulse On non-ejector items, five electrical releases shall be made with the minimum specified voltage applied for a 0 020 second impulse. Both the lightest and heaviest stores specified shall be released under each of the following item temperature stabilized conditions Each lag time shall be recorded

a. 93°C (200°F)

b Room temperature

c. -56 7°C (-70°F).

If the item has an ejector system, the above tests shall be performed by normal electrical ignition of the ejector system. Each lag time shall be recorded.

4.5.2.2 Dive and climb release On non-ejector items, one release shall be at the nominal specified voltage using both the lightest and heaviest stores specified, under each of the following item temperature stabilized conditions and in each of the specified attitudes. Each lag time shall be recorded.

a Temperature stabilized conditions.

1. 93°C (200°F).

2 Room temperature

3. -56.7°C (-70°F)

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b Attitudes for horizontal bombing items

1 15° climb

2 15° dive

c Attitudes for dive bombing items

1 15° climb

2 30° climb

3 90° climb

4 30° dive

5 70° dive

If the item has an ejector system, the above tests shall be performed by normal initiation of the ejector system Each lag time shall be recorded. In addition, the item's performance shall be determined by measuring the functional parameters in accordance with 3 6

4 5 2 3 Visual The item shall be visually inspected after each of the conditions above to ensure it has not been damaged due to the lag time determination tests

4 5 3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure

4 5 3 1 Horizontal release. The maximum lag time shall not exceed 0 100 second The item shall release the store and there shall be no damage to the item as a result of store release

4 5 3 2 Dive and climb release The maximum lag times shall not exceed the following

a	15° climb and dive	0 100 second
b.	30° climb and dive	0 100 second
c.	70° dive	0 150 second
d.	90° climb	0.400 second

The item shall release the store and there shall be no damage to the item as a result of store release.

4 5 3.3 Visual. The item shall not malfunction as a result of excessive wear (except replaceable parts), nor shall the testing cause decals, name-plates paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, or discolor There shall be no damage to the item as a result of store release.

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4 6 Icing test.

4.6.1 Conditions. The item shall be tested to ensure that it functions under icing conditions as required. The intent of the icing tests is to determine whether rapid change from altitude to sea level conditions or freezing rain/snow will cause equipment to malfunction. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3 7 The tests are divided into two categories: 1 - Navy and Air Force, 2 - Army

4.6.2 Procedure.

4 6 2.1 Navy and Air Force The primary inflight release method of the item shall be used while suspending the lightest and heaviest stores The lightest store shall be used during the first freeze (4 6.2.1.1) while the heaviest store shall be used during the second freeze (4 6 2 1 2) for primary releases In addition, each secondary or emergency inflight release method of the item shall be used while suspending the lightest and heaviest stores The lightest store shall be used during the first freeze, while the heaviest store shall be used during the second freeze for secondary releases. Ground service operation shall not be tested.

4 6 2 1 1 First freeze The item shall be exposed to the test environment in the loaded condition. All compatible arming wires shall be inserted in the arming units. If applicable, the electric fuze arming connector plug shall be inserted into the electric fuze arming unit. The temperature of the item shall be reduced to -56.7°C (-70°F), or the item shall be placed in a chamber at -56.7°C (-70°F) and the temperature stabilized The item shall be placed in an atmosphere having a temperature of 37.8°C (100°F) and a relative humidity of 90 percent until all evidence of frost has disappeared Retaining all the condensation practicable, the temperature of the item shall be reduced to -56.7°C (-70°F) or the item shall be placed in a chamber at -56.7°C (-70°F) and the temperature stabilized. The following tests shall be conducted:

a. Store release. The specified store, suspended from the item in a horizontal attitude, shall be released electrically at the minimum specified voltage with an impulse of 0.020 second duration. The lag time shall be recorded. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6.

b. Mechanical fuze arming, armed. With the mechanical arming unit electrically energized at minimum specified voltage, it shall be verified that each arming unit is ARMED by subjecting it to a pull of 68.04 +2.27, -0 kilograms (150 +5, -0 pounds) per figure 1, direction 1.

c. Mechanical fuze arming, safe setting. With the mechanical arming unit electrically SAFE, the force required to pull the arming wires out of each arming unit, per figure 1, direction 1, shall be measured.

d. Electric fuze arming The electric fuze arming unit shall be tested as specified.

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4 6.2.1 2 Second freeze The item shall be loaded with the required store, the arming wires and electric fuze arming connector plug inserted, and the item placed in an atmosphere having a temperature of 37.8°C (100°F) and a relative humidity of 90 percent, until all evidence of frost has disappeared. Retaining all the condensation, the temperature of the item shall be reduced to -56.7°C (-70°F) or the item shall be placed in a chamber set at -56.7°C and the temperature stabilized. The tests specified in 4 6 2.1.1 shall be repeated After all tests have been completed, the item shall be dried

4.6 2.2 Army The primary inflight release method of the item shall be tested while suspending the lightest and heaviest stores In addition, each secondary or emergency inflight release method of the item shall be tested while suspending the lightest and heaviest stores. Ground service operation shall not be tested

4.6 2.2.1 Freezing rain The item shall be exposed to the test environment in the loaded condition with the lightest store All compatible arming wires shall be inserted in the arming units The electric fuze arming connector shall be inserted into the electric fuze arming unit The temperature of the item shall be reduced to and stabilized at -17 8°C (0°F). After stabilization, the chamber temperature shall be increased to -6 7°C (20°F) The test item shall then be sprayed with water precooled to 1 7°C (35°F) at a rate of 2 54 centimeters (1 inch) per hour from all directions to cover the top and sides of the item. When at least 0.635 centimeter (one quarter inch) of glaze is accumulated on any part of the item, the tests of 4 6 2.1 1 shall be conducted If difficulties are encountered, the ice shall be broken away or melted using resources normally available to men in the field and the results shall be reported to the acquiring activity The test shall be repeated with the heaviest store and for secondary release methods

4.6.2.3 Visual The item shall be visually inspected after each of the conditions above to ensure the item has not been damaged due to the freezing or icing conditions.

4 6.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.6.3.1 Navy, Air Force, and Army The item, when tested as specified, shall meet the following criteria·

a. Store release. The lag time shall not exceed 0.100 second The item shall release the store. There shall be no damage to the item as a result of store release.

b. Operating forces, arming unit The arming unit shall hold the 68.04 kilogram (150 pound) proof load (see 4.3.2.2.2) without damage and shall function after load application. The arming unit shall release between the minimum and maximum loads for SAFE release specified without damage and shall function after any SAFE release If SAFE release loads are not available see 3.6.

c. Electric fuze arming. The electric fuze arming unit shall meet the requirements specified.

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4.6.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of the icing test or store release

4 7 Vibration test

4 7 1 Dynamic vibrational stresses A vibration test shall be performed to determine if the item is constructed to withstand expected dynamic vibrational stresses and to ensure that performance degradation or malfunctions will not be produced by the service vibration. The item shall be tested to ensure that it functions as required Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3 7. The tests shall be performed in each of three mutually perpendicular axes in any order. All testing shall be completed in one axis prior to testing in the next axis

4 7 2 Procedure.

4 7 2 1 Test item operation Unless otherwise specified, the test item shall be checked for electrical release (excluding explosives) and inflight manual operation during application of vibration (resonance search, resonance dwell, cycling, and random vibration), so that the effects caused by these tests shall be evaluated At the conclusion of each test axis, the test item shall again be checked for electrical release (excluding explosives) and inflight manual operation, and the results shall be compared with data obtained prior to the vibration testing. If the item has no method of inflight release except explosive ejection, the manual ground service release will be operated at the conclusion of each test axis and the results shall be compared with the requirements as specified or shall be in accordance with 3 6

4 7.2.1.1 Continuity. Each electrical circuit of the item shall be continuously monitored to check its continuity Inert cartridges or squibs shall be used in place of live ammunition. Special attention shall be given to switches, and intervalometers, in order to see if inadvertent stepping or chatter occurs.

4.7.2.1 2 Preloads. All sway braces and any structural part or other component normally used in service and capable of affecting the vibration environment shall be utilized, adjusted to the same preload levels as specified for operational employment

4.7.2.2 Mounting techniques. The test item shall be attached to the vibration exciter table by its normal in-service mounting, by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup Whenever possible, the item weight shall be distributed uniformly on the vibration exciter table in order to minimize effects of unbalanced loads The fixture shall be such that the transverse motion at the input monitoring point(s) shall be minimized, and should be limited to 100 percent of the input motion

4 7 2.2 1 Suitable test store. A suitable store is defined as one having a weight of 453 6 kilograms (1000 pounds) or greater, but not exceeding the heaviest store specified for the item. Items limited to stores of less than 453 6 kilograms (1000 pounds) shall be tested with their maximum weight store specified.

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4 7.2.2 2 Inert stores. The test loads should be applied by using service inert stores or missiles rather than simulated stores. If simulated stores must be used, the weight, center of gravity (CG), moment of inertia, and mounting interface shall be identical to the actual store, including structural rigidity.

4 7 2 3 Test requirements

4.7.2.3 1 Sinusoidal vibration Vibration shall be applied along each of the three mutually perpendicular axes of the test item. The vibratory acceleration levels or double amplitudes of the specified curve shall be maintained on the fixture at the test item support points. When specified, for sinusoidal resonance dwell and cycling tests of items weighing more than 36 28 kilograms (80 pounds) mounted in helicopters, the vibratory accelerations shall be reduced one g for each 9 07 kilogram (20 pound) increment over 36 28 kilograms (80 pounds). Acceleration derating shall apply to the highest test level of the selected curve, but in no case shall the derated test level be less than 50 percent of the selected curve

4.7.2.3 2 Random vibration, control Two control accelerometers shall be located on the fixture at the test item support points. Two control accelerometers shall be mounted on the item adjacent to the lowest interface with the suspended store. All control accelerometers shall be oriented with their sensitive axis parallel to the applied vibration The average of all four control accelerometers shall be equal to the specified test level Examples of control accelerometer location are as follows.

a Where a store suspension item is intended to be tested with a store, for both the unloaded and loaded test, two control accelerometers shall be mounted to the fixture adjacent to each mounting point The additional two control accelerometers shall be mounted on the item adjacent to the applicable suspension hooks. All accelerometer signals will be averaged for control of vibration.

b. Where a store suspension item mounts to another store suspension item, as a multiple ejection rack (MER) or triple ejection rack (TER) mounted to a bomb rack, two control accelerometers shall be mounted on the fixture as above, and the additional two shall be mounted on the frame of the lowest rack or launcher adjacent to the suspension points of the lowest store. Where there are two or more lowest racks as with the MER, one control accelerometer may be located on the front of the forward rack and one on the rear of the aft rack of the group, each adjacent to the suspension point All four accelerometers will be averaged for control.

4.7.2.3.3 Random vibration test. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level The power spectral density (PSD) of the test control signal shall not deviate from the specified requirements by more than +100, -50 percent (±3 decibels (dB)) between 20 Hz and 2000 Hz, except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz, maximum, between 500 and 2000 Hz.

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Tolerance levels in terms of dB are defined as

$$dB = 10 \log \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density in g^2/Hz units.

The term W_0 defines the specified level in g^2/Hz units.

Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product of BT = 50, minimum Specific analyzer characteristics shall be as specified below or equivalent

a On-line, contiguous filter, equalization/analysis system having a bandwidth B = 50 Hz, maximum

b Swept frequency analysis systems characterized as follows.

1 Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 20 Hz, maximum between 20 and 200 Hz

B = 50 Hz, maximum between 200 and 2000 Hz

(b) Analyzer averaging time = T = 2RC = 1 second, minimum, where T = true averaging time and RC = analyzer time constant.

(c) Analysis sweep rate (linear) = S = $\frac{B}{4RC}$ or $\frac{B^2}{8}$ Hz/sec, whichever is smaller

2. Constant percentage bandwidth analyzer

(a) Filter bandwidth = pf_c = one third octave minimum ($0.23f_c$) where p = percentage and f_c = analyzer center frequency.

(b) Analyzer averaging time = T = $\frac{50}{pf_c}$, minimum

(c) Analysis sweep rate (logarithmic) = S = $\frac{pf_c}{4RC}$, or $\frac{(pf_c)^2}{8}$ Hz/second, whichever is smaller

c. Digital power spectral density analysis system employing quantization techniques providing accuracies corresponding to the above approach.

The composite g-rms test levels shall not be less than the value specified. Accelerometers employed for test level control shall be mounted in accordance with 4 7.2 3 2.

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4 7 2 3 4 Combined temperature-vibration test Vibration testing shall be required at -56 7°C (-70°F) and 93°C (200°F) if specified All other procedures shall be the same as required by room temperature vibration

4 7.2 4 Procedure 1 – equipment installed on helicopters.

4 7.2 4 1 Sinusoidal vibration, search unloaded With the item in the released condition, it shall be vibrated for a maximum period of one-half hour, and the existence of all resonances or complex motions of the item or its components within the frequency range of 5 to 500 Hz shall be determined The speed of the sweep through the frequency range shall be one octave per minute or less The test levels shall be according to the specified curve on figure 2

4 7 2 4 2 Sinusoidal vibration, search loaded The item shall be loaded with its specified store of suitable weight and the test of 4 7 2.4.1 repeated.

4 7 2 4 3 Sinusoidal vibration, unlatched resonance dwell The test item, in the released condition, shall be vibrated along each axis at the resonant frequencies determined in 4.7 2.4 1 Each resonant frequency shall be vibrated for a period of 15 minutes The test levels shall be according to the specified curve on figure 2 If more than four significant resonant frequencies are found for any one axis, the four most severe resonant frequencies shall be chosen for the dwell test If a change in the resonant frequency occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition The final resonant frequency shall be recorded

4.7 2 4 4 Sinusoidal vibration, loaded resonance dwell The test of 4 7 2 4 3 shall be repeated at the resonant frequencies determined in 4 7 2 4 2, with a suitable test store installed

4.7 2 4 5 Release and transfer at resonance. A minimum of two manual or electrical (excluding explosives) releases shall be made at each of the loaded resonant conditions to ensure that the release and transfer mechanisms operate satisfactorily. The transfer switch shall be tested in accordance with 4.4.2.6. The electrical release shall be accomplished using nominal specified voltage impulses 0 020 second long with a dwell of 0.020 second between each impulse If no resonance was observed during the resonant search of 4.7 2 4 2, the two releases shall be made at a frequency of 50 Hz In addition, the item shall be subjected to a 5 minute period of vibration at each resonant point after which the vibration shall be stopped and a release accomplished to ensure that the vibration has not caused the item to jam.

4.7.2 4.6 Sinusoidal cycling vibration-unloaded. The test item, in the released condition, shall be vibrated along each axis for a period of one and one-half hours less the time spent at resonant dwells in 4 7.2.4.3 The frequency of applied vibration shall be swept at 1 octave per minute between 5 and 500 Hz at the test levels specified on figure 2

4.7 2 4 7 Sinusoidal cycling vibration-loaded The test item, with a suitable test store installed, shall be vibrated along each axis for a period of one and one-half hours less the time spent at resonant dwells in 4 7.2 4.3. The frequency of applied vibration shall be swept at 1 octave per minute between 5 and 500 Hz at the test levels specified on figure 2

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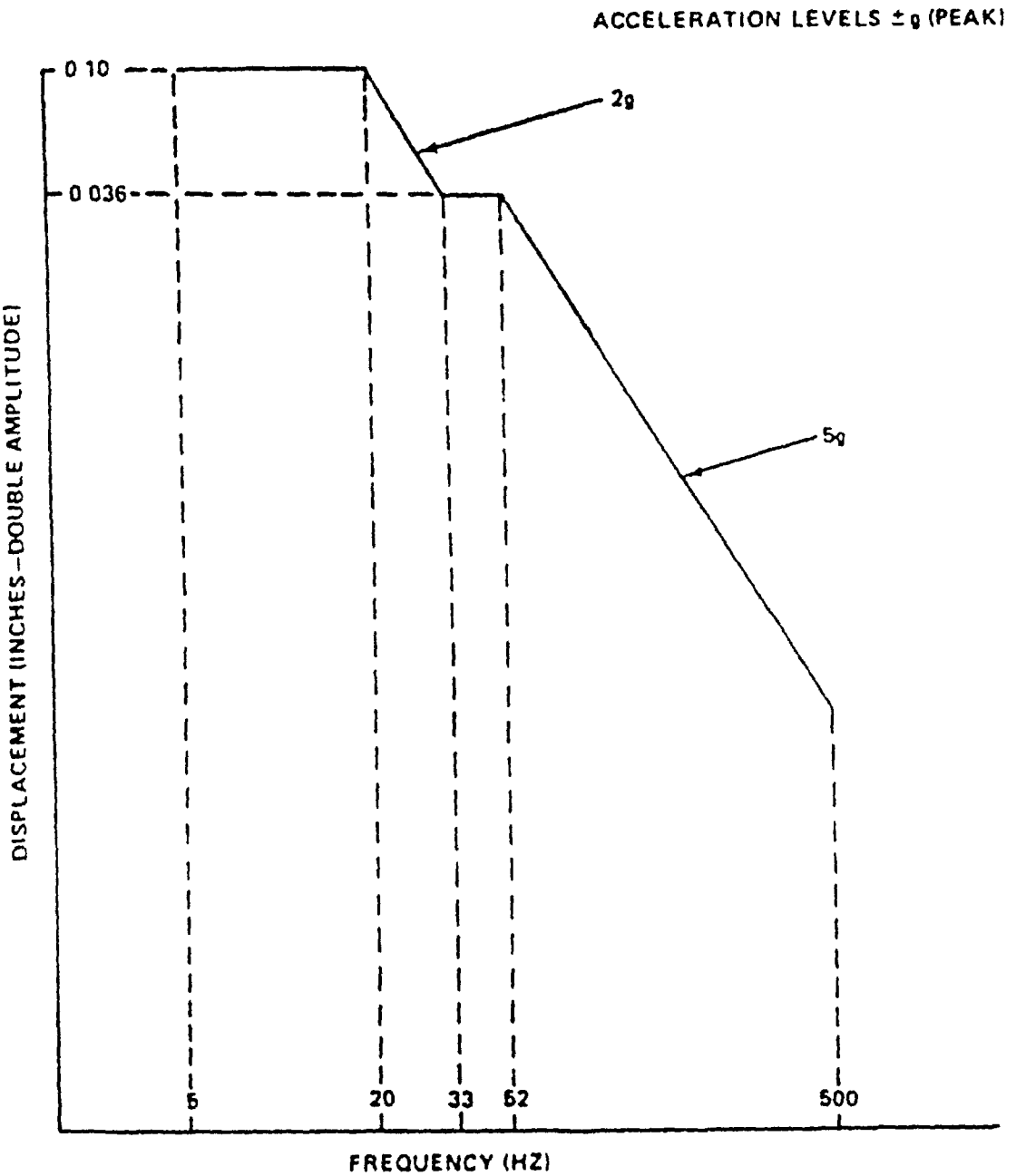


FIGURE 2. Vibration test curve for equipment installed in helicopters.

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4.7.2 5 Procedure II - equipment installed in aircraft other than helicopters

4 7 2 5.1 Random vibration-unloaded The test item, in the released condition, shall be vibrated for a minimum period of 40 minutes in each of three mutually perpendicular axes The test spectrum shall be flat in the range of 20 to 2000 Hz at a PSD (g^2/Hz) level of 0 02

4 7 2 5 2 Random vibration-loaded The test item shall be loaded with a suitable test store and sway braced as required for service installation The test of 4 7.2 5 1 shall then be repeated except that for each 45 35 kilogram (100 pound) increment of suspended load over 500 pounds, the acceleration spectral density may be reduced by 0 5 dB However, the specified curve may not be reduced by more than 2 5 dB as specified in table III

TABLE III Acceleration (g-rms) vs load

Suspended load (lbs)	Spectral 1/ density, g^2/Hz	Frequency range, Hz	Over acceleration, g-rms	Reduction, dB
Unloaded	0 02	20 - 2K	6 3	0
Up to 500	0 018	20 - 2K	5 9	0
500 - 600	0 016	20 - 2K	5 6	-1.0
600 - 700	0 014	20 - 2K	5 3	-1 5
700 - 800	0 0126	20 - 2K	5 0	-2 0
800 - 900	0 011	20 - 2K	4 7	-2 5
900 - 1000	0 011	20 - 2K	4 7	-2 5
Over 1000	0.011	20 - 2K	4 7	- 2.5

1/ Flat spectrum in the range of 20 to 2000 Hz

4 7.2.5 3 Release and transfer A minimum of two manual or electrical (excluding explosives) releases (see 4 7 2 1) shall be made during the final 5 minutes of loaded vibration to ensure that the releases and transfer switch operate satisfactorily The transfer switch shall be tested in accordance with 4 4 2.6. The electrical release shall be accomplished using nominal specified voltage impulses 0 020 second long with a dwell of 0 020 second between each impulse In addition, the item shall be subjected to a 5 minute period of loaded vibration after which a release shall be accomplished to ensure that the vibration has not caused the item to jam

4 7 2 6 Functional. After the completion of vibration in all test axes, the item shall be loaded with the lightest store The store shall be released by the primary release means In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3 6 The test shall be repeated with the heaviest store

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4.7.2.7 Visual. The item shall be visually inspected after each of the conditions above to ensure the item has not been damaged due to vibration. In addition, the contractor shall supply written instructions which cover the disassembly and inspection, after completion of all vibration tests, of the component parts of the item by non-destructive means; i e., magnaflux, dye penetrant, x-ray, and dimensional measurements.

4 7.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.7.3.1 Vibration. At the conclusion of each test axis or any portion thereof, the item shall release and arm as specified. Inadvertent release of the store during vibration or failure to release the store during vibration or after vibration for any cause shall not occur. Structural damage to the item shall not occur during vibration or during store release. Malfunction of the electrical circuits (open circuits, switch chatter, and improper resistance) detected during the continuity checks shall not occur

4 7.3 2 Functional. The item shall release the store and there shall be no damage to the item as a result of store release. The data measured shall conform to the item's detailed specification or shall be in accordance with 3 6

4 7.3 3 Visual Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of vibration testing or store release. Disassembly of the item (see 4.7.2.7) after the completion of all vibration tests shall show no damage or deformation such as cracks, permanently set components, warpage, and excessive wear

4.8 Shock test.

4.8.1 Support points. The shock test is performed to determine if the item is constructed to withstand expected shock stresses. Unless otherwise specified, the support points shall be subjected to the required acceleration in both directions along the three principal axes of the item. The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3.7. If catapult or arrested landing load tests are not applicable to the item, the tests of 4.8.2.7 shall be omitted.

4.8.2 Procedure.

4.8.2.1 Test item operation. At the conclusion of each test axis, the test item shall be checked for electrical release (excluding explosives) and inflight manual operation and the results compared with data obtained prior to the shock testing. If the item has no method of inflight release except explosive ejection, the manual ground service release will be performed at the conclusion of each test axis and the results shall be compared with the requirements of the item's detailed specification or shall be in accordance with 3 6.

4.8.2.2 Mounting techniques The test item shall be attached to the shock table by its normal in-service mounting by means of a rigid fixture

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capable of transmitting the shock conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible, the item weight shall be distributed uniformly on the shock table to minimize effects of unbalanced loads. Shock amplitudes and durations shall be measured by techniques that will not significantly affect test item input control or response

4.8.2.2.1 Preloads All sway braces and any structural part or other component normally used in service and capable of affecting the shock environment shall be utilized, and shall be adjusted to the same preload levels as specified for operational employment

4 8 2 2 2 Inert stores Test loads shall be applied using service weight inert stores or missiles rather than simulated stores. If simulated stores must be used, the weight, CG, moment of inertia and mounting interface shall be identical to the actual store, including structural rigidity.

4 8 2.3 Common test techniques In each of the following tests, the item shall be subjected to the acceleration encountered when the support points are subjected to the specified acceleration when the item is mounted as in 4 7 2 2. The peak applied values of the accelerations shall be within ±5 percent of the values specified below and shall reach peak value in a time specified in 4.8 2 4 through 4 8 2 7 The waveform of acceleration shall resemble a half sine wave

4 8 2.4 No load The item shall be subjected to an acceleration of 25 g's while supporting no load in the latched condition The acceleration shall reach peak value in not more than 0 60 second and not less than 0.015 second

4.8 2.5 Light load. The test specified in 4.8.2 4 shall be repeated with the item supporting the lightest store specified If the acceleration of 25 g's results in a dynamic loading which is in excess of the permissible yield load for the item, the maximum value of acceleration specified shall be used. The acceleration shall reach peak value in not more than 0.60 second and not less than 0.015 second.

4.8.2.6 Yield load. Supporting the maximum weight store specified, the item shall be tested to the accelerations specified but not to exceed the yield. Therefore, when testing to yield loads, the resultant store acceleration shall be within +0 and -10 percent of the specified yield load in that direction. More than one shock may be applied in order to approach the specified levels from the low side. The time to reach peak value shall not be more than 0.60 second and not less than 0.015 second.

4.8.2.7 Catapult and arrested landing load. For items intended for catapult and arrested landings aboard aircraft carriers, the tests of 4.8.2.5 and 4.8.2.6 shall be repeated in both the forward and aft directions. The acceleration shall reach peak value in not more than 0.75 second and not less than 0.4 second.

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4.8.2.8 Functional. After the completion of shock testing in all test axes, the item shall be loaded with the lightest store The store shall be released by the primary release means. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6. The test shall be repeated with the heaviest store.

4.8 2.9 Visual. The item shall be visually inspected after each of the conditions above to ensure the item has not been damaged due to shock. In addition, there shall be written instructions (see 6.3) which cover the disassembly and inspection, after completion of all shock tests, of the component parts of the item by non-destructive means; i e , magnaflux, dye penetrant, x-ray, and dimensional measurements

4 8.3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.8 3.1 Shock, light and yield load. At the conclusion of each axis of testing, or any portion thereof, the item shall release and arm as specified Inadvertent release of the store during shock or failure to release the store after shock for any cause shall not occur Structural damage to the item or store release during shock testing shall not occur

4 8 3 2 Functional The item shall release the store and there shall be no damage to the item as a result of store release The data measured shall conform to the item's detailed specification or shall be in accordance with 3.6.

4.8 3.3 Visual Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of shock testing or store release. Disassembly of the item (see 4.8 2 9) after the completion of all tests shall show no damage or deformation such as cracks, permanently set components, warpage, and excessive wear.

4.9 High-g release test.

4.9.1 Accelerations. The item shall be tested to ensure it functions when subjected to loads created by accelerations which can be experienced as a result of aircraft maneuvers. It shall be tested to ensure that it functions as required. In addition, the item's performance shall meet the appropriate functional parameters in accordance with 3 6.

4.9.2 Procedure.

4.9.2.1 High-g release. Unless otherwise specified, the item shall release a load of 7 1/2 times the weight of the heaviest store up to 1587.6 kilograms (3500 pounds) or a load of 5 1/2 times the heaviest store over 1587.6 kilograms (3500 pounds) in a direction vertically downward as follows:

a. Twenty-five consecutive primary releases at rated voltage shall be performed. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6.

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b Three consecutive releases shall be performed on each additional secondary or emergency inflight release method provided.

NOTE

The 25 primary releases shall be considered as part of heavy store releases during the life test of 4.10

4 9.2.2 Visual. The item shall be visually inspected after each of the conditions above to ensure the items have not been damaged due to high-g release. In addition, there shall be written instructions (see 6.3) which cover the disassembly and inspection, after completion of all high-g tests of the component parts of the item by non-destructive means, i e , magnaflux, dye penetrant, x-ray, and dimensional measurements

4 9 3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4 9 3 1 High-g release The item shall release and arm as specified The item shall not inadvertently release the store Structural damage to the item shall not occur during load application or store release.

4 9.3.2 Visual Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of the high-g application store release. Disassembly of the item (see 4 9.2 2) after the completion of all tests shall show no damage or deformation, such as cracks, permanently set components warpage, and excessive wear

4 10 Life test.

4.10.1 Construction. The life test is performed to determine if the unit is constructed to withstand an extended number of releases. The number of releases is not to be considered indicative of the design or service life of the item. Normal cleaning and maintenance procedures shall be performed as required. However, cleaning and maintenance procedures described in service manuals do not apply except as directed by the acquiring activity. For explosive items, when a bleed orifice is used, it shall be replaced when its diameter exceeds the specified limit or has increased 10 percent over the original diameter. In addition, any maintenance operation (not previously required) found during testing that is required to keep the unit operational will be reported to the acquiring activity The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3 7

4 10.2 Procedure

4.10.2.1 Nonejector item. Five hundred electrical releases of the item shall be made at room temperature and at nominal specified voltage, using the heaviest store specified. This test shall be repeated for the lightest store specified. The time interval between releases shall be sufficient to prevent overheating of the electrical release unit The cumulative releases conducted under all tests shall not exceed the total specified under life test The lag

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times shall be recorded for the following releases.

a. 1st to 10th release, inclusive.

b. 121st to 130th release, inclusive

c. 246th to 255th release, inclusive.

d. 371st to 380th release, inclusive

e. 491st to 500th release, inclusive

Each secondary or emergency inflight method of release shall be tested for five releases using the lightest store specified These releases shall not be counted toward life test releases.

4.10.2.2 Ejector item. For those items with an ejector system, the test specified in 4.10 2.1 shall be conducted, except that 125 ejections using the lightest store and 125 ejections using the heaviest store shall be made The lag times shall be recorded for the following releases

a. 1st to 10th release, inclusive.

b. 51st to 60th release, inclusive

c. 116th to 125th release, inclusive

Each secondary or emergency inflight method of release shall be tested for five releases using the lightest store specified These releases shall not be counted toward life test releases.

4.10 2.2.1 Instrumented ejections Five instrumented ejections, as a minimum, to determine ejector and reaction force, store acceleration peak value, and terminal ejection velocity (see 3.6), will be included in the first 10 ejections and five more included in the last 10 ejections of each store weight tested, except for secondary or emergency releases.

4.10.2.2.2 Maintenance demonstration for ejector items. For items required to demonstrate 100 ejections over a 30 day period without ejector system cleaning/maintenance, the following test shall be conducted The test shall be counted towards life test releases. Further, the test shall be conducted using only one store weight and, for items cartridge powered, using the largest (propellant weight) impulse cartridges specified. The test shall be conducted to the following criteria:

a. The item shall be completely serviced and cleaned prior to testing.

NO MAINTENANCE DURING TEST

b. At least 20 of the 100 ejections shall be instrumented. However, the first 3 and last 3 ejections must be instrumented, and the item shall have a consistency of performance of ±5 percent (see 3.6) throughout the test.

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c. During the first 20 days of testing, 95 ejections shall be completed. No more than 10 ejections can be done during any one day.

d. The item shall not be operated from day 21 through day 29. On day 30, the last five ejections shall be completed.

4.10.2.3 Visual. The item shall be visually inspected after each of the conditions above to ensure the items have not been damaged due to life testing. In addition, there shall be written instructions (see 6.3) which cover the disassembly and inspection, after completion of all life tests, of the component parts of the item by non-destructive means, i.e., magnaflux, dye penetrant, x-ray, and dimensional measurements. If specified, a complete dimensional analysis of component parts shall be made to determine the amount of wear.

4.10.3 Failure criteria. If the item does not meet the specified/established performance criteria or the following criteria, it shall be cause for failure.

4.10.3.1 Life test. The item shall release the store either mechanically or at required forces, velocities, accelerations, and times as specified. The acquiring activity reserves the right to make measurements on any release or ejection to establish this fact. Damage to the item caused by the power cartridge and test fixture, shall be reported to the acquiring activity. The acquiring activity will advise whether the test item will be repaired or replaced. If repaired, testing shall be continued from the point of failure. The maximum allowable lag time shall be 0.100 second. If the item specification has no data requirements, there shall be no significant difference in the first and last data measurements of each weight class (see 3.6).

4.10.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of the life testing. Disassembly of the item (see 4.10.2.3) after completion of all tests shall show no damage or deformation such as cracks, permanently set components, and warpage. The dimensional analysis of the components shall be reported to the acquiring activity.

4.11 Ejector system.

4.11.1 Overpressure. The ejector system test is performed to determine if the system is constructed to withstand a significant overpressure which could result from blocked pressure relief ports, and lock shut firing. The item shall be tested to ensure that it functions as required.

4.11.2 Procedure.

4.11.2.1 Ejector system. The entire ejector system shall be hydrostatically pressurized to two and one-half times the maximum operating pressure (excluding lock shut firing) for a period of 1 minute. However, components can be removed, such as telescoping pistons and foot assemblies, that cannot be readily sealed to hold pressure. All fixed, pressurized components shall be tested. Plugs or fixtures, approved by the acquiring activity (see 3.7), shall be installed to replace any removed part.

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4.11.3 Failure criteria If the item does not meet the following criteria, it shall be cause for failure.

4.11.3 1 Ejector system The ejector system shall hold the applied pressure for the 1 minute period. Warpage, bending, buckling, and yielding of the ejector system are allowed as long as pressure is maintained for the 1 minute period.

4.12 Sand test.

4.12.1 Sand-laden atmosphere. The sand test is performed to determine if the item functions when subjected to a fine, sand-laden atmosphere. All electrical connectors and normally closed orifices shall be suitably protected only to the extent that the mating connectors and hardware mate satisfactorily The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3 7.

4.12 2 Procedure

4 12.2 1 Dust requirements. The dust used in this test shall be fine sand (97 - 99 percent SiO_2 by weight) of angular structure, and shall have the following size distribution as determined by weight, using the U.S Standard Sieve Series.

a. 100 percent of this dust shall pass through a 100-mesh screen

b 98 ± 2 percent of the dust shall pass through a 140-mesh screen

c 90 ± 2 percent of the dust shall pass through a 200-mesh screen

d 75 ± 2 percent of the dust shall pass through a 325-mesh screen

4.12.2.2 Test item. Place the test item in the chamber, positioned as near the center as possible The item shall be tested with latches or hooks open. Any part or components of the item (such as extended ejection pistons) that would normally be exposed to the atmosphere when a store is not installed, shall be exposed to the test environment. No surface of the test item shall be closer than 4 inches from any wall of the test chamber. Orient the item so as to expose the most critical or vulnerable parts to the dust stream. The test item orientation may be changed during the test if so required by the equipment specification. Parts of the item normally protected when installed on the aircraft shall be adequately covered during the test. The chamber shall maintain a temperature of 23°C (73°F) and a relative humidity of less than 22 percent. The air velocity shall be 533.4 ± 76.2 meters (1750 ± 250 feet) per minute. The dust concentration shall be 0.3 ± 0.2 grams per 28.32 liters (1 cubic foot). With test item not operating, maintain these conditions for 6 hours. The dust-laden air shall be introduced into the chamber in such a manner so that its flow is approximately laminar before it strikes the test item. At the completion of the test allow the test item to return to ambient conditions.

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4 12 2.3 Post-sand tests After removal from the test chamber, the item shall be subjected to the following tests:

a. Mechanical fuze arming, SAFE setting. With the mechanical arming unit electrically SAFE, the force required to pull all compatible arming wires out of the arming units per figure 1, direction 1 shall be measured.

b Mechanical fuze arming, ARMED. With the mechanical arming unit energized at nominal specified voltage, it shall be verified that each arming unit is ARMED by subjecting each to a pull of 68.04 + 2.27, -0 kilograms (150 +5, -0 pounds) per figure 1, direction 1.

c Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage The lag time shall be measured In addition, the item's performance shall be determined by measuring the functional parameters in accordance with 3 6

d Electric fuze arming The electric fuze arming unit snall be tested as specified

4.12 2.4 Visual. The item shall be visually inspected after removal from the test chamber for damage caused by the sand test After arming and store release, the item shall again be inspected to ensure that it has not been damaged due to the arming or release tests

4.12.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.12 3 1 Post-sand tests.

a. Operating forces, arming units. The arming unit shall hold the 68.04 kilogram (150 pound) proof load (armed) without any damage. It shall function after load application. The arming unit shall release between the minimum and maximum loads for SAFE release specified in its detailed specification without any damage It shall function after any SAFE release. See 3.6 if SAFE release loads are not available.

b. Store release. The lag time shall not exceed 0.100 second. The item shall release the store. There shall be no damage to the item as a result of store release.

c. Electric fuze arming The electric fuze arming unit shall meet the requirements specified

4.12.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of the sand test or store release.

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4.13 Salt spray.

4.13.1 Salt atmosphere. The salt spray is performed to determine if the item functions when subjected to a salt atmosphere. All electrical connectors and normally closed orifices shall be protected only to the extent that the mating connectors and hardware mate satisfactorily. The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3.7.

4.13.2 Procedure.

4.13.2.1 Salt spray test. The item shall be subjected to the salt spray test of ASTM B 117. If no time is designated in the item's detailed specification, 120 hours shall be used. The item shall be tested with latches or hooks open. Any part or component of the item (such as extended pistons) that would normally be exposed to the atmosphere when a store is not installed, shall be exposed to the test environment.

4.13.2.2 Post-salt tests. The operational test at rated voltage of 4.4.2.2 shall be performed at the end of the spraying period and within 1 hour. The item shall then be loaded with the lightest store and allowed to dry for 48 hours. The item shall not be operated in any way during the drying period. At the end of the drying period, the following tests shall be performed.

a. Store release. The lightest store shall be released by the primary means with the minimum specified voltage applied for a 0.020 second impulse. The lag time shall be measured. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6. If failure of the electrical connector is observed, it shall be cleaned of salt or moisture and the test repeated from the point of failure.

b. Secondary release. Using the lightest store, the item shall be reloaded and each secondary or emergency inflight release shall be tested.

c. Mechanical fuze arming, ARMED. With the mechanical arming unit electrically energized at the minimum specified voltage, it shall be verified that each arming unit is ARMED by subjecting it to a pull of 68.04 +2.27, -0 kilograms (150 +5, -0 pounds) per figure 1, direction 1.

d. Mechanical fuze arming, SAFE setting. With the mechanical arming unit electrically SAFE, the force required to pull the arming wires out of the arming units per figure 1, direction 1, shall be measured.

e. Electric fuze arming. The electric fuze arming unit shall be tested as specified.

4.13.2.3 Visual. The item shall be visually inspected after the above conditions to ensure the parts have not been damaged due to the salt spray test or store release.

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4.13.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.13.3.1 Post-salt test.

a. Operational test at rated voltage Each circuit shall operate in the specified manner when actuated with the specified voltage (see 4.4.3.2) The item shall hold the store after this test.

b. Store release. The maximum lag time shall not exceed 0 100 second. The item shall release the store as specified There shall be no damage to the item as a result of store release.

c. Secondary release. The item shall release the store and there shall be no damage to the item as a result of store release. The operating forces or distances shall meet the requirements of the item or shall be in accordance with 3 6

d Operating forces, arming unit The arming unit shall hold the 68 04 kilogram (150 pound) proof load without any damage. It shall function after load application The arming unit shall release between the minimum and maximum loads for SAFE release specified without any damage and shall function after any SAFE release If SAFE release loads are not available see 3 6

4.13 3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of salt spray or store release Corrosion products are acceptable as long as they do not impair the function of the item However, any corrosion noted shall be examined and reported to the acquiring activity

4.14 Stress corrosion.

4.14.1 Stress corrosion cracking. The stress corrosion test is performed on items that are subjected to a salt or corrosive environment while in a loaded condition for long periods of time. Examples of this would be a bomb rack with a sway braced fuel tank held for months at a time. The purpose of the test is to determine if the item is susceptible to stress corrosion cracking. The salt spray test of 4.13 may replace this test when approved by the acquiring activity. The item shall be tested to ensure that it functions as required.

4.14.2 Procedure

4.14.2.1 Preparation. The item shall be subjected to all static test limit loads as specified. The loads shall be applied per 4.16 (static tests)

4.14.2.2 Stress corrosion. The item shall be placed in a salt spray chamber and tested as required in 4.13.1 and 4.13.2.1, except that 80 percent of the greatest design limit load in the vertical down direction shall be applied. At the completion of the time specified in 4.13.2.1, the load shall be reduced to zero and the item shall be removed from the chamber. The post-salt tests of 4 13 2 2 shall then be accomplished The item will then be

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placed back in the chamber, with the load reapplied, until the total chamber time equals 336 hours. The item shall be inspected daily to ensure that the applied load has caused no structural failures.

4.14.2.3 Post-stress corrosion tests.

a. Visual. After 336 hours, the item will be removed from the chamber and allowed to dry for 48 hours. It shall then be inspected and all discrepancies and corrosion products shall be noted.

b. Store release. At the end of the drying period, the item shall be loaded with the heaviest store specified and a primary release made at rated voltage. If failure of the electrical connector is observed, it shall be cleaned of salt and moisture and the test continued from the point of failure.

4.14.3 Failure criteria. If the item does not meet the following criteria, it shall be cause for failure.

4.14.3.1 Stress corrosion. The item shall meet the failure criteria of 4.13.3 when removed from the test chamber after the time specified in 4.13.2.1. During the remainder of the test, the item shall hold the applied load. All observations made during the visual inspection shall be reported to the acquiring activity.

4.14.3.2 Store release. The item shall release the store. Cracks, warpage, and yielding of the item are acceptable as long as the store is released.

4.15 Salt spray/altitude

4.15.1 Hermetically sealed assemblies. The purpose of this test is to verify the adequacy of the hermetic seal on any component parts of the test item. Since testing at sea level pressures may not cause leakage at pinholes, the combination salt spray/altitude conditions will cause a salt laden atmosphere to be drawn into any sealed units if a small leak exists. Therefore, the test shall only be performed on those items that have hermetically sealed assemblies which can be affected by the salt/altitude combination and shall be at the discretion of the acquiring activity. The item shall be tested to ensure that it functions as required or as directed by the acquiring activity.

4.15.2 Procedure.

4.15.2.1 Salt spray/altitude. The item, in an unloaded condition, shall be subjected to the salt spray test of 4.13.2.1 except the duration shall be 24 hours. Immediately upon removal from the salt spray chamber, the item shall be placed in an altitude chamber at room temperature and subjected to the pressure at 21,336 ± 1067 meters (70,000 ± 3500 feet) for 1 hour. The item shall be operated in accordance with 4.4.2.2 while at test altitude. After the 1 hour period, the chamber pressure shall be reduced to sea level as fast as possible, but in less than 5 minutes. While the pressure is being reduced and for 5 minutes after reaching sea level, the item shall be sprayed with a salt water solution prepared in accordance with 4.13.2.1. The salt solution shall be sprayed only in those areas with hermetically sealed parts. The spray shall be sufficient to wet the surface of the test item and the

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relative humidity shall be 85 percent or better at the spray nozzle, when measured at sea level The pressure chamber shall be maintained at room temperature throughout the test At the end of the spray period, the item shall be operated in accordance with 4 4 2 2

4 15 2 2 Visual The item shall be visually inspected after the above conditions to ensure the parts have not been damaged due to the salt spray/altitude test. After this test, the item may be washed and cleaned prior to further testing

4 15 3 Failure criteria If the item does not meet the following criteria, it shall be cause for failure

4 15 3 1 Salt spray/altitude At the completion of the salt spray/altitude test, the item shall be operated as specified and shall conform to the failure criteria of 4.4.3 2

4.15 3 2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of salt spray/altitude testing

4.16 Static tests.

4.16 1 Critical design conditions Static test of the item shall be conducted for all critical design conditions for the item The item shall support the maximum weight store specified, or a dummy store with identical geometry in the region having contact with the item and its sway braces Tests shall be conducted for each combined load and moment condition in such a manner that the loads are applied in increments not greater than 10 percent of the limit load Features/mechanisms of the item shall meet the requirements of 3 4, 3 6, and 3 7

4.16 2 Procedure

4.16.2 1 Design limit loads. The item shall be tested to all conditions of design limit loads and moments as defined in 6.4.10. Each condition (set of loads) shall be simultaneously applied as specified The item shall be subjected to the following tests after application and removal of all limit load conditions.

a. Operating forces, manual release mechanism The item shall be loaded with the heaviest store and the store shall be released. Operating forces shall be measured in accordance with 4.3.2.1.

b. Store release. The heaviest store shall be suspended from the item and released by the primary means at rated voltage. The lag time shall be measured. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6.

4.16.2.2 Yield loads. The item shall be tested to the yield loads and moments as defined in 6.4.22. Each condition (set of loads) shall be simultaneously applied as specified The item shall be subjected to the tests of 4 16 2 1 after application and removal of all yield load conditions.

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4.16.2.3 Visual. After completion of design limit and yield load test and prior to application of ultimate loads, the item shall be visually inspected to ensure the items have not been damaged due to static load application In addition, there shall be written instructions (see 6.3) which cover the disassembly and inspection, after completion of all static load tests, of the component parts of the item by non-destructive means; i.e., magnaflux, dye penetrant, x-ray, and dimensional measurements.

4.16.2.4 Ultimate loads. The item shall be tested to all conditions of ultimate loads and moments as defined in 6 4.21. Each condition (set of loads) shall be simultaneously applied as specified All limit and yield load tests shall be completed before any ultimate conditions are applied

4.16.3 Failure criteria. If the item does not meet the specified/ established performance criteria (except for ultimate load) or the following criteria, it shall be cause for failure.

4.16 3.1 Design limit loads. Application of the design limit loads shall result in no permanent deformation of the item. In addition, the item shall meet the following requirements·

a Operating forces, manual release mechanism. The operating forces or distances shall meet the requirements of the item's detailed specification or shall be in accordance with 3 6. There shall be no damage to the item as a result of store release

b Store release. The lag time shall not exceed 0 100 second The item shall release the store. There shall be no damage to the item as a result of store release.

4.16.3.2 Yield loads. Permanent deformation may remain after application and removal of any yield load. However, it shall not adversely affect the aerodynamic characteristics of the functional operation of the aircraft or store installation, nor shall such deformation be so great that it can be detected without dimensional analysis. In addition, the item shall meet the criteria of 4.16.3.1.

4.16.3.3 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of static load testing (limit and yield) or store release. Disassembly of the item (see 4.16.2.3) after completion of limit and yield loads shall show no damage such as cracks, warpage, and excessive wear (see 4.16.3.2).

4.16.3.4 Ultimate loads. The store shall be retained such that failure shall be defined as the inability of the item to hold applied loads or retain the store upon application of ultimate loads. Deformation or structural breakage of the item does not constitute failure at the ultimate load level

4.17 Rain.

4.17.1 Normal service conditions. The rain test is conducted to determine the effectiveness of items which are exposed to rain under normal service conditions. All electrical connectors and normally closed orifices shall be protected only to the extent that the mating connectors and hardware mate satisfactorily The item shall be tested to ensure that it functions as

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required. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3.7.

4.17 2 Procedure.

4.17 2.1 Chamber temperature. The chamber temperature shall be uncontrolled, except as regulated by water introduced as rain, throughout the test period. The rain shall be produced by a water distribution device of such design that the water is emitted in the form of droplets having a diameter range between 1 and 4 millimeters The temperature of the water shall be between 11° and 35°C (52° and 95°F) The wind source shall be so positioned with respect to the equipment that it will cause the rain to beat directly and uniformly against one side of the equipment The wind source shall be capable of producing horizontal wind velocity up to 64.36 kilometers (40 miles) per hour. The wind velocity shall be measured at the position of the test item, prior to placement of the item in the chamber No rust or corrosive contaminants shall be imposed on the test item by the test facility

4.17 2 2 Test item The test item shall be placed in the chamber in its normal operating position The item shall be tested with latches or hooks open Any parts or components of the item (such as extended ejection pistons) that would be exposed to the atmosphere when a store is not installed shall be exposed to the test environment The test item shall be exposed to a simulated rain at a rate of 12 7 ± 2 54 centimeters (5 ± 1 inches) per hour for 10 minutes The rate of rainfall shall then be raised to 30 48 ± 2 54 centimeters (12 ± 1 inches) per hour and held at this rate for 5 ± 1 minutes The rate shall then be reduced to 12.7 ± 2 54 centimeters (5 ± 1 inches) per hour for the next 15 ± 1 minutes. Starting 5 ± 1 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 64.36 kilometers (40 miles) per hour The wind source shall be maintained at this velocity for 15 minutes, after which it shall be turned off. Each of the sides of the test item that could be exposed to blown rain shall be subjected to the rain for a period of not less than 30 minutes, for a total duration of not less than 2 hours.

4.17 2 3 Post-rain tests. After removal from the test chamber, the item shall be subjected to the following tests

a. Operational test at rated voltage. To ensure that the item has been properly assembled, all circuits shall be operational when nominal specified voltage is applied to the item terminal(s) or connector(s). The test shall be conducted in accordance with 4.4 2.2.

b. Store release. The lightest store shall be suspended from the item and released by the primary means, nominal specified voltage. The lag time shall be measured In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6.

c Electrical fuze arming. The electric fuze arming unit shall be tested as specified.

4.17 2.4 Visual. The item shall be visually inspected after removal from the test chamber for damage caused by the rain test The item shall be inspected to ensure that it has not been damaged due to the release

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4.17 3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4 17 3.1 Post-rain tests.

a. Operational test at rated voltage. Each circuit shall operate in the specified manner when actuated with nominal specified voltage. The failure criteria of 4.4.3.2 shall apply

b. Store release. The lag time shall not exceed 0 100 second The item shall release the store. There shall be no damage to the item as a result of store release.

c. Electric fuze arming The electric fuze arming unit shall conform to the requirements as specified

4.17 3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of the rain testing or store release

4.18 Humidity.

4.18.1 Warm, highly humid atmosphere The humidity test is conducted to determine the resistance of an item to a warm, highly humid atmosphere. All electrical connectors and normally closed orifices shall be protected only to the extent that the mating connectors and hardware mate satisfactorily. The item shall be tested to ensure that it functions as required Features/mechanisms of the item shall meet the requirements of 3.4, 3 6, and 3.7.

4.18.2 Procedure.

4 18.2.1 Determination of relative humidity. The chamber and accessories shall be constructed and arranged in such a manner as to avoid condensate dripping on the test item. The chamber shall be trap-vented to the atmosphere to prevent the buildup of total pressure Relative humidity shall be determined from the dry bulb-wet bulb thermometer comparison method or an equivalent method. When readout charts are used, they shall be capable of being read with a resolution within 0.6°C (1°F). When the wet bulb control method is used, the wet bulb and tank shall be cleaned and a new wick installed at least every 30 days. The air velocity flowing across the wet bulb shall be not less than 274.3 meters (900 feet) per minute. Provisions shall be made for controlling the flow of air throughout the internal test chamber area where the velocity of air shall not exceed 45 7 meters (150 feet) per minute. Steam, or distilled, demineralized, or deionized water having a pH value between 6.0 and 7.2 at 23°C (73°F) shall be used to obtain the specified humidity. No rust or corrosive contaminants shall be imposed on the test item by the test facility.

4 18 2.2 Test item The test item shall be placed in the test chamber in its normal operating position. The item shall be tested with latches or hooks open. Any parts or components of the item (such as extended ejection pistons) that would normally be exposed to the atmosphere when a store is not installed shall be exposed to the test environment Prior to starting the test, the internal chamber temperature shall be at standard ambient with

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uncontrolled humidity. Gradually raise internal chamber temperature to 65°C (149°F) and the relative humidity to 95 percent over a period of 2 hours Maintain these conditions for not less than 6 hours. Maintaining 85 percent, or greater, relative humidity, reduce the internal chamber temperature in 16 hours to 30°C (86°F). Repeat this procedure for 10 cycles, until the total chamber time is not less than 240 hours

4.18.2.3 Post-humidity tests. After removal from the test chamber, the item shall be subjected to the following tests

a Operational test at rated voltage To ensure that the item has been properly assembled, all circuits shall be operational when nominal specified voltage is applied to the item terminal(s) or connector(s) The test shall be conducted in accordance with 4.4 2 2

b Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage The lag time shall be measured In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3 6

c Electric fuze arming The electric fuze arming unit shall be tested as specified

4.18.2 4 Visual The item shall be visually inspected after removal from the test chamber for damage caused by the humidity test The item shall be inspected to ensure that it has not been damaged due to the store release

4.18 3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure

4.18.3.1 Post-humidity tests.

a. Operational test at rated voltage. Each circuit shall operate in the specified manner when actuated with nominal specified voltage The failure criteria of 4.4.3.2 shall apply

b. Store release. The lag time shall not exceed 0.100 second. The item shall release the store There shall be no damage to the item as a result of store release.

c. Electric fuze arming. The electric fuze arming unit shall conform to the requirements as specified.

4.18.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of the humidity testing or store release.

4.19 Fungus.

4.19.1 Resistance to fungi. The fungus test is conducted to determine the resistance of an item to fungi and the conditions most favorable for fungi development, namely high humidity, warm atmosphere and presence of inorganic salts All electrical connectors and normally closed orifices shall be

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protected only to the extent that the mating connectors and hardware mate satisfactorily. The item shall be tested to ensure that it functions as required Features/mechanisms of the item shall meet the requirements of 3 4, 3.6, and 3.7.

4.19.2 Procedure.

4.19.2.1 Preparation of fungi. The following methods shall be used when preparing the fungi for test. The item shall be tested with latches or hooks open

4.19.2.1.1 Equipment. The apparatus required to conduct this test consists of chambers or cabinets together with auxiliary instrumentation capable of maintaining the specified condition of temperature and humidity. Provisions shall be made to prevent condensation from dripping on the test item. There shall be free circulation of air around the test item and the surface area of fixtures supporting the test item shall be kept to a minimum.

4 19.2.1.2 Preparation of mineral-salts solution The solution shall contain the following:

Potassium dihydrogen orthophosphate (KH_2PO_4)	0 7 gram
Potassium monohydrogen orthophosphate (K_2HPO_4)	0 7 gram
Magnesium sulfate ($MgSO_4 \cdot 7H_2O$)	0 7 gram
Ammonium nitrate (NH_4HO_3)	1 0 gram
Sodium chloride (NaCl)	0.005 gram
Ferrous sulfate ($FeSO_4$ $7H_2O$)	0.002 gram
Zinc sulfate ($ZnSO_4 \cdot 7H_2O$)	0.002 gram
Manganous sulfate ($MnSO_4$ $7H_2O$)	0.001 gram
Distilled water	1000 ml

Sterilize the mineral salts solution by autoclaving at 121°C (150°F) for 20 minutes. Adjust the pH of the solution by the addition of 0.01 normal solution of NaOH so that after sterilization, the pH is between 6 0 and 6 5. Prepare sufficient salts solution for the required tests.

4.19.2.1.3 Purity of reagents Reagent grade chemicals shall be used in all tests. Unless otherwise specified, it is intended that all reagents shall conform to the specification of the Committee on Analytical Reagents of the American Chemical Society, where such specifications are available.

4.19.2.1.4 Purity of water. Unless otherwise specified, reference to water shall be understood to mean distilled water or water of equal purity.

4.19.2.1.5 Preparation of mixed spore suspension. The following test fungi shall be used:

Fungi	ATTCC 1/ no.	NLBSS 2/ no.
Aspergillus niger	9642	386
Aspergillus flavus	9643	380
Aspergillus versicolor	11730	432
Penicillium funiculosum	9644	391
Chaetomium globosum	6205	459

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1/ American Type Culture Collection, 12301 Parklawn Drive, Rockville, MD 20852

2/ Pioneering Research Div , U S. Army Natick Laboratories, Natick, MA 01760

Maintain culture of these fungi separately on an appropriate medium such as potato dextrose agar. However, the culture of Chaetomium globosum shall be cultured on strips of filter paper on the surface of mineral-salts agar (Mineral-salts agar is identical to mineral-salts solution described in 4.19.2 1 2 but contains in addition 15.0 grams of agar per liter) The stock cultures may be kept for not more than 4 months at 6°C (43°F) Use subcultures incubated at 29°C (84°F) for 7 to 20 days in preparing the spore suspension. Prepare a spore suspension of each of the five fungi by pouring into one subculture of each fungus a sterile 10 ml portion of water or a sterile solution containing 0.05 grams per liter of a nontoxic wetting agent such as sodium dioctyl sulfosuccinate Use a sterile platinum or nichrome inoculating wire to gently scrape the surface growth from the culture of the test organism Pour the spore charge into a sterile 125 ml glass-stoppered Erlenmeyer flask containing 45 ml of sterile water and 10 to 15 solid glass beads, 5 mm in diameter Shake the flask vigorously to liberate the spores from the fruiting bodies and to break the spore clumps Filter the shaken suspension through a thin layer of sterile glass wool in a glass funnel into a sterile flask in order to remove mycelial fragments Centrifuge the filtered spore suspension aseptically, and discard the supernatant liquid Resuspend the residue in 50 ml of sterile water and centrifuge Wash the spores obtained from each of the fungi in this manner three times. Dilute the final washed residue with sterile mineral salts solution in such a manner that the resultant spore suspension shall contain 1,000,000 ± 200,000 spores per ml as determined with a counting chamber. Repeat this operation for each organism used in the test and blend equal volumes of the resultant spore suspension to obtain the final mixed spore suspension. The spore suspension may be prepared fresh each day or may be held at 6°C (43°F) for not more than four days.

4.19.2 1.6 Viability of inoculum control With each daily group of tests, place each of three pieces of sterilized filter paper, 1 inch square, on hardened mineral salts agar in separate petri dishes. Inoculate these with spore suspension by spraying the suspension from a sterilized atomizer (Devilbiss No. 154 atomizer or equivalent has been found satisfactory for this purpose) so that the entire surface is moistened with the spore suspension. Incubate these at 29°C (84°F) at a relative humidity not less than 85 percent and examine them after 14 days incubation There shall be copious growth on all three of the filter paper control specimens. Absence of such growth requires repetition of the test.

4.19.2.1.7 Control items. In addition to the viability of inoculum control, a number of known susceptible substrates shall be inoculated along with a test item to ensure that proper conditions are present in the incubation chamber to promote fungus growth The control items shall include three pieces each of preservative-free, vegetable-tanned leather and protein-glue bonded cork.

4 19.2 2 Test item.

4 19.2 2 1 Inoculation of test and control items. Mount the test and control items on fixtures or suspended from hangers Next, pre- condition the chamber and its contents at 29°C (84°F) and 95 percent relative humidity for

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at least 4 hours. Inoculate the test and control items with the mixed fungus spore suspension by spraying it on the test and control items in the fine mist from a previously sterilized atomizer or nebulizer until both are thoroughly wet with the spray. Incubation is to be started immediately following the inoculation.

4.19.2.2 2 Incubation. Maintain the test chamber at 29°C (84°F) and 95 percent relative humidity, minimum, during the life of the test Keep the test chamber closed during the incubation period except during inspection or for addition of other test items. After 14 days, inspect the control items. They should show an abundant growth of fungus. If the control items do not show an abundant growth, the entire test shall be repeated If the control items show satisfactory fungus growth, continue the test for a period of 28 days from the time of inoculation

4.19 2.2.3 Post-fungus tests. After removal from the test chamber the item shall be subjected to the following tests:

a. Operational test at rated voltage. To ensure that the item has been properly assembled, all circuits shall be operational when nominal specified voltage is applied to the item terminal(s) or connector(s) The test shall be conducted in accordance with 4.4.2 2

b Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage. The lag time shall be measured In addition, the item's performance shall be determined by measuring the functional parameters in accordance with 3.6

c. Electric fuze arming. The electric fuze arming unit shall be tested as specified.

4.19.2.2.4 Visual. The item shall be visually inspected after removal from the test chamber for damage caused by the fungus test. The item shall be inspected to ensure that it has not been damaged due to the store release.

4 19.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.19.3.1 Post-fungus tests.

a. Operational test at rated voltage. Each circuit shall operate in the specified manner when actuated with nominal specified voltage. The failure criteria of 4.4.3.2 shall apply.

b. Store release. The lag time shall not exceed 0.100 second. The item shall release the store. There shall be no damage to the item as a result of store release.

c. Electric fuze arming. The electric fuze arming unit shall conform to the requirements as specified.

4 19.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts to peel, chip, loosen, break, or

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discolor. There shall be no damage to the item as a result of the fungus testing or store release.

4.20 Altitude.

4.20 1 Effects of reduced pressure The altitude test is conducted to determine the effects of reduced pressure on the item Damaging effects may include leakage of gases from gasket sealed enclosures, rupture of pressurized containers and erratic operation of electrical components due to arcing or corona. If the salt spray/altitude test of 4.15 has been previously performed, this test may be deleted. The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3.4, 3 6, and 3 7

4 20 2 Procedure.

4 20 2.1 Test item. The item shall be tested with latches or hooks open. Any parts or components of the item (such as pistons that hang down) that would be exposed to the atmosphere when a store is not installed, shall be exposed to the test environment The internal chamber temperature shall be reduced to -54°C (-65°F) The internal pressure of the chamber shall be reduced to 21,336 ± 1067 meters (70,000 ± 3500 feet) The equipment shall be maintained under these conditions for 1 hour or as specified At the conclusion of this period and while test altitude and temperature conditions are maintained, the equipment shall be operationally checked within 1 hour Electrical checks shall be conducted as specified by the acquiring activity. Ejection is not required at altitude conditions. As a minimum, the electrical operation test of 4 4 2 2 shall be performed and all electrical devices operated.

4 20.2 2 Post-altitude tests After removal from the test chamber, the item shall be subjected to the following tests

a. Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage. The lag time shall be measured. In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3.6.

b Electric fuze arming. The electric fuze arming unit shall be tested as specified

4.20.2.3 Visual The item shall be visually inspected after removal from the test chamber for damage caused by the altitude test. The item shall be inspected to ensure that it has not been damaged due to the store release

4.20.3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.20.3.1 Chamber tests, operational test at rated voltage. Each circuit shall operate in the specified manner when actuated with nominal specified voltage. Inability of the item to load and hold the lightest store after this test shall be considered a failure

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4 20.3.2 Post-altitude tests

a Store release. The lag time shall not exceed 0.100 second. The item shall release the store. There shall be no damage to the item as a result of store release.

b Electric fuze arming The electric fuze arming unit shall conform to the requirements as specified.

4.20.3 3 Visual Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of the altitude testing or store release

4.21 Sunshine.

4 21 1 Effect of solar radiation The sunshine test is conducted to determine the effect of solar radiation on items. Items with parts made of rubber, cloth, and plastic, are generally those affected by this type of test The item shall be tested to ensure that it functions as required Features/mechanisms of the item shall meet the requirements of 3 4 3 6, and 3 7

4 21 2 Procedure

4 21.2 1 Radiation source The test chamber volume shall be a minimum of ten times that of the envelope volume of the test item. The chamber's simulated solar radiation source area shall be a minimum of 125 percent of the horizontal area projection of the test item. For the purposes of this test, the following spectral distribution of solar radiation is acceptable. 538 to 775 watts per square meter (50 to 72 watts per square foot) of infrared (wavelengths above 7800 angstrom units), 43 to 75 watts per square meter (4 to 7 watts per square foot) of ultraviolet (wavelengths below 3800 angstrom units), and the balance visible The radiation source shall be located at least 76.2 centimeters (30 inches) away from any outer surface of the test item. (Lamp vendor's spectral distribution curves may be used in establishing the spectral distribution within the above specified limits U.S. Bureau of Standards traceability of this vendor data is waived.) Tests which are conducted for degradation and deterioration of materials may use one of the following acceptable radiation sources: (1) mercury vapor lamps (internal reflector type only), (2) combination of incandescent spot lamps (including infrared filters) together with tubular type mercury vapor lamps with external reflector, (3) combination of incandescent spot lamps (including infrared filters) together with mercury vapor lamps with internal reflectors (with filters as required), (4) carbon arc lamps with suitable reflectors, or (5) mercury-xenon arc lamps with suitable reflectors (with filters as required).

4.21.2.2 Test item. The test item shall be placed in the test chamber in its normal operating position. The item shall be tested with latches or hooks open. Any parts or components of the item (such as pistons that hang down) that would be exposed to the atmosphere when a store is not installed shall be exposed to the test environment Expose unit to radiant energy at the rate of 1076 to 1292 watts per square meter (100 to 120 watts per square foot) or as specified in the equipment specification. The period of the test shall not be less than 48 hours, during which time the chamber temperature shall be maintained at 45°C (113°F).

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4 21 2 3 Post-sunshine tests After removal from the test chamber, the item shall be subjected to the following tests

a. Operational test at rated voltage To ensure that the item has been properly assembled, all circuits shall be operational when nominal specified voltage is applied to the item terminal(s) or connector(s) The test shall be conducted in accordance with 4.4 2 2

b. Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage The lag time shall be measured In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3 6

c Electric fuze arming The electric fuze arming unit shall be tested as specified

4 21 2.4 Visual. The item shall be visually inspected after removal from the test chamber for damage caused by the sunshine test The test shall be inspected to ensure that it has not been damaged due to the store release

4.21 3 Failure criteria If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure

4.21 3 1 Post-sunshine tests

a Operational test at rated voltage Each circuit shall operate in the specified manner when actuated with nominal specified voltage. The failure criteria of 4.4.3 2 shall apply.

b. Store release. The lag time shall not exceed 0.100 second The item shall release the store There shall be no damage to the item as a result of store release

c. Electric fuze arming. The electric fuze arming unit shall conform to the requirements as specified

4.21.3.2 Visual Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor. There shall be no damage to the item as a result of the sunshine testing or store release

4.22 Temperature shock.

4.22.1 Sudden changes in temperature. The temperature shock test is performed to determine the effects on items of sudden changes in the temperature of the surrounding atmosphere The item shall be tested to ensure that it functions as required. Features/mechanisms of the item shall meet the requirements of 3.4, 3.6, and 3.7

4.22.2 Procedure

4.22 2.1 Test item The equipment to be tested shall first be placed in a test chamber in its normal operating position where a temperature of 93°C (200°F) is maintained The item shall be tested with latches or hooks open

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Any part or component of the item (such as extended ejection pistons) that would be exposed to the atmosphere when a store is not installed, shall be exposed to the test environment. The equipment shall be exposed to this temperature for a period of 4 hours, at the conclusion of which, and within 5 minutes, the equipment shall be transferred to a chamber having an internal temperature of -54°C (-65°F) The equipment shall be subjected to this temperature for a period of 4 hours This constitutes one cycle The number of complete cycles shall be three. The duration of exposure at each extreme temperature shall not be less than specified and may be extended to overnight exposure to prevent interruption of the transfer sequence. At the conclusion of the third cycle, the equipment shall be removed from the test chamber

4 22.2 2 Post-temperature shock tests. After removal from the test chamber, the item shall be subjected to the following tests.

a Operational test at rated voltage To ensure that the item has been properly assembled, all circuits shall be operational when nominal specified voltage is applied to the item terminal(s) or connector(s). The test shall be conducted in accordance with 4 4 2 2

b Store release. The lightest store shall be suspended from the item and released by the primary means at nominal specified voltage The lag time shall be measured In addition, the item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3 6

c Electric fuze arming The electric fuze arming unit shall be tested as specified

4 22 2 3 Visual. The item shall be visually inspected after removal from the test chamber for damage caused by the temperature shock test. The item shall be inspected to ensure that it has not been damaged due to the store release

4 22.3 Failure criteria. If the item does not meet the specified/ established performance criteria or the following criteria, it shall be cause for failure.

4.22.3.1 Post-temperature shock tests.

a. Operational test at rated voltage. Each circuit shall operate in the specified manner when actuated with nominal specified voltage The failure criteria of 4.4.3.2 shall apply

b. Store release. The lag time shall not exceed 0.100 second. The item shall release the store There shall be no damage to the item as a result of store release.

c. Electric fuze arming. The electric fuze arming unit shall conform to the requirements as specified.

4.22.3.2 Visual. Testing shall not cause decals, nameplates, paint, plating, cotter pins, snap rings, and nuts, to peel, chip, loosen, break, or discolor There shall be no damage to the item as a result of the temperature shock testing or store release

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4 23 Lock shut firing

4.23.1 Adequacy of safety devices. Lock shut firing is conducted to ascertain the item's safety devices to retain the store in the event of an inadvertent electrical or mechanical attempt to release. The item shall be tested to ensure that it functions as required.

4.23.2 Procedure.

4 23.2.1 Lock shut firing. The item stabilized at 93.3°C (200°F), loaded with the heaviest store, shall be subjected to a lock shut firing as follows

a Items incorporating electrical safing devices An attempt shall be made to fire the item, using the primary release system with the safing devices installed or placed in a safe position The unit shall not fire

b Items incorporating mechanical safing devices The item shall be fired, using the primary release system, with the safing device installed or placed in a safe position

c Items incorporating electrical and mechanical safing devices Both the electrical and mechanical safing devices shall be installed or placed in a safe position An attempt shall be made to fire the item using the primary release system The unit shall not fire The electrical interlock shall then be bypassed or armed and the item fired with the mechanical device safed

d. As a minimum, the ejector piston forces shall be measured

4.23 2 2 Visual The item shall be visually inspected after the above tests to ensure that the store has not been released due to the lock shut firing.

4.23.3 Failure criteria. If the item does not meet the following criteria, it shall be cause for failure. There shall be no evidence of broken parts.

4.23.3.1 Lock shut firing electrical devices. The unit shall not fire or release the store

4.23.3.2 Lock shut firing mechanical devices. The unit shall not fire or release the store with the electrical and mechanical safeties in the safe position. However, once the electrical safety is bypassed or armed, release of the store shattering or breakage of parts shall not occur when the unit fires with the mechanical safety in the safe position. Deformation of the mechanical safing device shall be permitted

4.24 Electromagnetic interference. The item shall be subjected to EMI, EMV, Hero, and EMP tests in accordance with the following applicable documents: MIL-STD-461C, Class A1b, test methods CE01, CE03/07, CS01, CS02, CS06, RE01, RE02, RS01, and RS02; MIL-STD-461C, test method RS03, 200 volt per meter test level for Air Force and Navy equipments and subsystems installed in non-metallic aircraft, non-metallic structures on metallic aircraft, or externally mounted on metallic aircraft; MIL-E-6051, MIL-HDBK-235 (parts 1, 2,

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and 3); and MIL-STD-1385. All test methods and failure criteria shall be contingent upon approval prior to the start of testing (see 3.7).

4.25 Sympathetic ignition

4 25 1 Ejector system. Sympathetic ignition is conducted only for ejector items powered by at least two impulse cartridges The test is to ascertain if the item's ejector system is properly designed such that one cartridge can ignite a second cartridge to effect safe store release. This is necessary if a cartridge has a faulty primer or if the electrical circuit to a cartridge fails in an open mode The item shall be tested to ensure that it functions as required.

4 25.2 Procedure

4 25.2.1 Sympathetic ignition The item, stabilized at -56.7°C (-70°F), loaded with the lightest store, shall be subjected to a sympathetic ignition ejection as follows:

a The item shall be powered with the smallest (propellant weight) impulse cartridges specified

b Nominal specified voltage shall be applied to the item's electrical connector such that only one cartridge's electrical circuit is powered to effect store release

c The item's performance shall be determined by measuring the appropriate functional parameters in accordance with 3 6.

4.25.2 2 Visual. The item shall be visually inspected after the above tests to ensure that the store has been released and that there is no damage to the item as a result of the store release.

4 25.3 Failure criteria If the item does not meet the following criteria, it shall be cause for failure There shall be no evidence of damaged or broken parts.

4.25 3.1 Sympathetic ignition. The store shall be released in a normal manner without damaging the item. All cartridges shall have ignited. The item's measured performance (which will be less than the base line data determined under 3.6) shall meet the criteria established by the acquiring activity.

4.26 Contamination. The item shall be subjected to contamination agents such as oil, grease, hydraulic fluid, jet fuel, de-icing fluid, and water/detergent solution to determine any detrimental effects. The contaminants shall be defined by the acquiring activity and shall be applied to the item as directed by the acquiring activity. Further, the item shall be mounted as if on an aircraft with mating electrical connectors installed. For example, an item may be used in-service in an aircraft pylon. Thus, for test purposes, the item shall be installed in a simulated pylon so that contaminants are applied to simulate normal service conditions. All test methods and failure criteria shall be contingent upon approval prior to the start of testing (see 3.7).

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4.27 Chemical hardening and decontamination The item, if accessible to exposure, shall be subjected to decontamination agents to determine any detrimental effects that could result from neutralizing various chemical/biological/radiological agents The decontaminants shall be defined and applied to the item as directed by the acquiring activity Further, the item shall be mounted as if on an aircraft with mating electrical connectors installed. For example, an item may be used in-service in an aircraft pylon. Thus, for test purposes, the item shall be installed in a simulated pylon so that decontaminants are applied to simulate normal service conditions. All test methods and failure criteria shall be contingent upon approval prior to the start of testing (see 3 7).

5 PACKAGING

This section is not applicable to this specification.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6 1 Intended use The tests of this specification are for use n determining reliability, suitability, and performance of store suspension equipment used in military aircraft

NOTICE When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person, or corporation, or conveying any rights, or permission to manufacture, use, or sell any patented invention that may in any way be related thereto

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

6.3 Data requirements. The following Data Item Descriptions (DIDs) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.5, 4.8.2 9, 4.9 2.2, 4.10 2 3, 4.16.2.3	DI-NDTI-80809A	Test/inspection report	---
3.7	DI-NDTI-80603	Test procedures	---
4 1.1	DI-NDTI-80809A	Test/inspection report	10 2 7, only

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423

6 4 Definition of terms

6.4.1 Arming. The process of arming prepares the fuze for detonation of the weapon. On the ground and in flight prior to release, the weapon is carried with its fuze safe, which prevents accidental detonation. The fuze arming is initiated during or subsequent to release by either mechanical or electrical methods, depending upon the type of fuze used

6 4 2 Bomb ejector rack A bomb ejector rack is an item which incorporates a method of forcefully separating the store from the aircraft at the time of release

6 4.3 Bomb rack. A bomb rack is an item rigidly attached to the aircraft structure for suspending and releasing stores Normally termed parent rack it may also be used to suspend multiple bomb racks, missile launchers, or rocket pods

6 4 4 Bomb shackle A bomb shackle is an item which may be nonrigidly attached to the aircraft and is primarily intended for use in bomb bays that employ vertical suspension on side rails. Thus, upon release of its store, the lower shackle swings out of the way, providing clearance for the store above it.

6 4 5 Ejection. Ejection is the process of propelling stores or equipment essentially normal to the aircraft line of flight, primarily to accomplish satisfactory separation.

6 4.6 Electric fuzes. Electrical fuze arming is accomplished by means of a breakaway expendable cable between the suspension item and the store The character of an electrical pulse transmitted to an electric fuze in the store at the time of release determines whether the store is dropped, armed, or safe

6.4.7 Lag time.

a. Nonejection items. The lag time for nonejection items is the interval between the beginning of the electrical release impulse to the item and the last contact between the store and such parts of the item as sway braces and the suspension hooks.

b. Ejector items

1 Air Force The lag time is the interval between the beginning of the rise of the force or pressure curve to the separation of the store from the item. Separation is the breaking of the last contact between the store and such parts of the item as sway braces and the suspension hooks when the ejection foot is insulated from the store

2. Navy and Army. The lag time is the interval between the beginning of the electrical release pulse to the separation of the store from the item Separation is the breaking of the last contact between the store

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and such parts of the item as sway braces and suspension hooks when the ejector foot is insulated from the store.

6.4.8 Launching. Launching is the operation of the suspension item so that the missile or rocket leaves the aircraft (generally in the direction of flight).

6.4.9 Light and heavy stores. The lightest and heaviest store of an item, as referred to in the text of this specification, shall be equivalent to the lightest and heaviest service stores used with that item as specified by the acquiring activity.

6.4.10 Limit load. Limit load is the maximum load anticipated in normal conditions of operation. The load shall be as specified in the detailed specifications and drawings for the item.

6.4.11 Manual function.

6.4.11.1 Inflight operation. Items containing provisions for remote manual operation by means of suitable controls in the aircraft. Manual operation may apply to either the release or the arming function. The normal usage of this system is for emergency or jettison release use.

6.4.11.2 Ground service operation. An item shall contain provisions for its manual operation by use of a specified release means on the item. The normal usage of this system is for routine ground servicing functions.

6.4.12 Mechanical fuzes. The typical mechanical fuze is essentially a vane driven fuze in the store which arms itself after the vane has made a prescribed number of revolutions. While carried in the safe condition, one end of an arming wire is attached to the vane, which prevents it from turning, and the other end is secured to the arming mechanism in the suspension item. Upon release, the store will drop armed if the arming wires are retained by the arming mechanism. An item may contain arming mechanisms, either integral or attachable, which provide for arming the tail and nose fuzes of the store.

6.4.13 Missile launcher. A missile launcher is an item attached to the aircraft to carry and launch, jettison, or eject a missile.

6.4.14 Release mechanism. A release mechanism is a device which allows complete separation of a store from a suspension item by operation of suspension hooks or other components normally used for suspending stores.

6.4.15 Retroejection. Retroejection is the ejection of a store in a direction opposite to the aircraft line of flight.

6.4.16 Secondary release. Secondary release is the emergency provision for clearing of stores or suspension items, or both, from the aircraft.

6.4.17 Stores. All missiles, rockets, bombs, mines, torpedoes, pyrotechnic devices, detachable fuel and spray tanks, pods, targets, and all similar items intended for carriage internally or externally by aircraft. This definition applies whether the items are, or are not, to be suspended from the aircraft in flight.

6.4.18 Support points. The support points are defined as the normal points attaching the item directly to the aircraft. If the item is not

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directly attached to the aircraft but utilizes auxiliary equipment in normal service use, such as a pylon, adapter, or racks, the support points shall be considered to be the attachment points of the auxiliary equipment to the aircraft

6.4.19 Terminal velocity. Terminal velocity is defined as the instantaneous velocity of the store when it leaves the foot of an item that has an ejector system.

6.4.20 Transfer switch. A transfer switch in an item is used to transmit the release signal to the next item when the items are used in a series, such as shackles in a bomb bay This ensures that the item is isolated from the next signal in the series

6.4 21 Ultimate factor of safety The minimum ultimate factor of safety shall be 1 5 times the limit load.

6.4 22 Yield factor of safety The minimum yield factor of safety shall be 1 15 times the limit load

6 5 Subject term (key word) listing

Ejector system
Icing test
Incubation mixed spore suspension
Manual function tests
Rain test
Salt spray
Sinusoidal sine

6.6 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Custodians:
Navy - AS
Air Force - 18

Preparing Activity
Navy - AS

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NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

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